

INTERNSHIP REPORT

*Comparison of Helixer Gene Predictions
Against Expert-Curated Annotations
in Plant Genomes of *

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GLOSSARY

UTR Non-coding mRNA region flanking the CDS; 5' UTR upstream of the start codon, 3' UTR downstream of the stop codon.

CDS Coding DNA Sequence: mRNA region translated into protein, bounded by start and stop codons.

Transcript / Isoform RNA product of a gene; alternative splicing may produce multiple isoforms per locus.

Ortholog Gene derived from a common ancestor through speciation; typically retains equivalent function across species.

Ab initio Gene prediction based solely on DNA sequence, without RNA-seq or homology evidence.

TEs Transposable Elements: DNA sequences capable of relocating within a genome; classified as Class I (retrotransposons) or Class II (DNA transposons).

ARAPORT11 *Arabidopsis thaliana* genome annotation (2017).

TAIR12 *Arabidopsis thaliana* genome annotation (2026).

Helixer Deep learning-based eukaryotic gene prediction tool.

GBOT Genome Browser and Orthologs Tracker: web platform for multi-species genome visualization and orthology exploration.

HelixFlor Python/R/SQL framework developed in this study for comparing Helixer predictions against reference annotations.

LaMME Laboratoire de Mathématiques et Modélisation d'Évry.

INRAE Institut national de recherche pour l'agriculture, l'alimentation et l'environnement.

IPS2 Institut des Sciences des Plantes de Paris-Saclay.

Comparison of Helixer Gene Predictions Against Expert-Curated Annotations in Plant Genomes of GBOT (Genome Browser & Ortholog Tracker).

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ABSTRACT

Accurate gene annotation remains challenging in plant genomes due to their size, repeat content, and structural complexity. Within the GBOT (Genome Browser and Orthologs Tracker) framework, we evaluated Helixer, a deep learning-based gene prediction tool, by extending a previous benchmark to two additional plant genomes, *Zea mays* and *Arabidopsis lyrata*, and by assessing the impact of reference annotation choice in *Arabidopsis thaliana* (ARAPORT11 vs. TAIR12). Using a comparison framework, Helixer predictions were evaluated against expert-curated annotations at the exon, coding sequence (CDS), and untranslated region (UTR) levels. Across eight plant genomes, Helixer showed strong structural agreement with reference annotations, with most species exceeding 70% shared gene models, although performance varied with genome complexity. In *A. lyrata*, Helixer reconstructed conserved gene structures despite the absence of this species in its training data, while functional support for novel predictions relied heavily on similarity to *A. thaliana* proteins. In *Zea mays*, a subset of Helixer-specific predictions overlapped annotated transposable elements, although this overlap was not significantly associated with the absence of protein-level functional evidence. Comparison of ARAPORT11 and TAIR12 showed that part of the variability attributed to prediction errors reflects inconsistencies between reference annotation versions. UTR boundaries were consistently less concordant with reference annotations than CDS regions across all species, indicating a recurring limitation of *ab initio* prediction. Overall, Helixer provides a scalable first-pass annotation approach for plant genomes, while interpretation of its predictions depends on genome complexity and reference annotation quality.

Keywords: Gene prediction, Helixer, structural annotation, transposable elements, comparative genomics.

INTRODUCTION

High-throughput sequencing has revolutionized our ability to read and study genomes. As a result, the number of available plant genome sequences has grown rapidly. Today, thousands of plant genomes are publicly available¹, including those of model plants like *Arabidopsis thaliana* and important crops such as rice, maize, and wheat. One of the main goals of *L'Institut national de recherche pour l'agriculture, l'alimentation et l'environnement* (INRAE), the French national research institute dedicated to agriculture, food, and the environment, is to understand how plants grow, adapt, and respond to their environment. Within INRAE, *L'Institut des Sciences des Plantes de Paris-Saclay* (IPSE) studies the genetic and molecular mechanisms that control plant growth, using a multidisciplinary approach that combines genomics, bioinformatics, and genetics. Therefore, it is essential to correctly identify and annotate all the elements present in these genomes.

Structural annotation is a critical step because errors at this stage can affect all downstream analyses, such as gene expression studies or evolutionary comparisons. Despite many advances, gene prediction remains difficult, especially in plant genomes. These genomes are often large and complex. They contain a high proportion of transposable elements (TE)², many repeated sequences, and duplicated genes. In addition, some plant species have undergone polyploidy events, meaning that they possess multiple complete sets of chromosomes, which may originate from the same or different species. All these characteristics make it difficult to correctly identify gene boundaries and structures. As a result, automatic gene prediction methods can produce errors such as missing genes, fragmented genes, or incorrect exon-intron structures. Traditionally, genome annotation relies on expert curation. This approach combines computational predictions with experimental data, such as RNA sequencing, and similarity with known genes from other species. Tools such as Augustus³ are commonly used to perform *ab initio* gene prediction. These predictions are then refined using additional evidence and manual curation by experts. Although this method can produce high-quality annotations, it is slow, expensive, and requires significant expertise. Consequently, many genomes, especially in non-model species, remain poorly annotated and have to be corrected, as is the case for *Arabidopsis lyrata*⁴. Indeed, it is a sister species of the model species *Arabidopsis thaliana* while remaining less extensively studied and annotated, it represents an interesting case for evaluating and improving automatic gene prediction methods through comparative analyses.

To overcome these limitations, new approaches based on machine learning and deep learning have been developed. These methods aim to learn patterns directly from genomic sequences without relying heavily on external data. Tools such as Tiberius⁵ and Helixer⁶ represent this new generation of gene prediction methods. They use neural networks trained on large datasets to identify gene structures directly from DNA sequences. Helixer, in particular, is a deep learning-based tool designed for eukaryotic gene prediction. It uses neural network models to detect coding regions and gene structures in genomic sequences. One of its main advantages is that it can work directly from raw DNA sequences, without requiring extensive additional data, especially useful for newly sequenced genomes or species with limited

experimental data. However, despite these advantages, the performance of Helixer is not always consistent across species. Its accuracy can depend on factors such as genome size, complexity, and similarity to the data used during training. Therefore, it is important to evaluate its predictions carefully by comparing them to high-quality reference annotations.

The Genome Browser and Orthologs Tracker (GBOT)⁷ provides a useful framework for this type of analysis. GBOT is a platform that integrates genomic data from different organisms, including bacteria, *Drosophila* species, and plants. It allows users to visualize genome annotations and to perform comparative analyses. It also includes tools to explore orthologous genes between species (**Figure 1**), which is important for genome annotation because conserved genes are more likely to be functional and biologically relevant. The identification of orthologs can therefore help validate predicted genes, transfer functional annotations, and detect potential annotation errors. In a previous study conducted within the GBOT framework, Helixer was applied to four plant genomes among the six in the database.

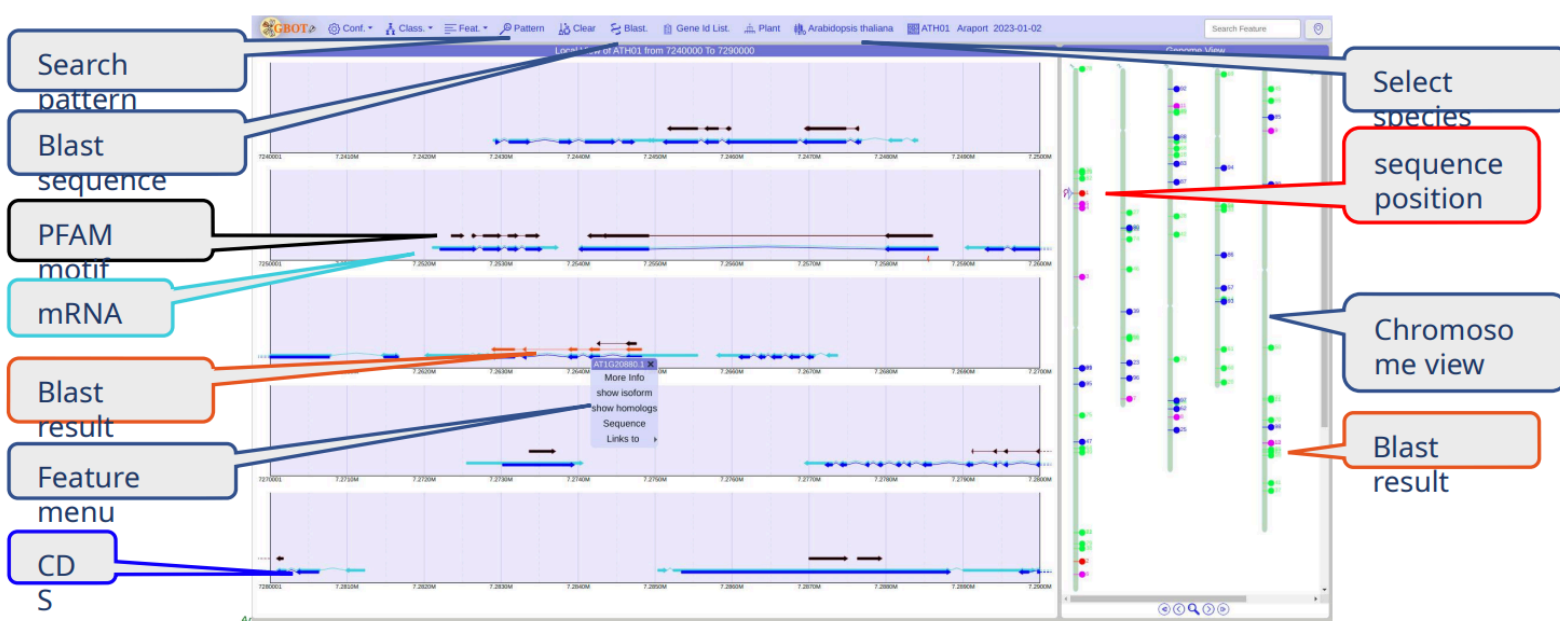


Figure 1: GBOT Web Interface

GBOT includes multiple tools dedicated to genome visualization, annotation comparison, sequence analysis, and comparative genomics across species :

- Search pattern:** Allows you to search for a specific sequence in the genome.
- Blast sequence:** Launches a BLAST search based on a given sequence.
- PFAM motif:** Displays the protein domains detected in the genes.
- mRNA:** Visualises the annotated mRNAs in the selected region.
- Blast result:** Shows the aligned results of the BLAST search.
- Feature menu:** Provides access to the various available annotations.
- CDS:** Displays the coding sequences of the genes.
- Select species:** Choose the species to display for comparison.
- Sequence position:** Indicates the exact position of the genomic sequences.
- Chromosome view:** Displays chromosome alignments between species.
- BLAST result (comparative):** Shows BLAST alignments between homologous genes from different species.

The predicted annotations were compared with expert-curated annotations in order to evaluate the performance of the tool. This study showed that Helixer has good potential for gene prediction, but also highlighted some limitations, such as errors in gene structure or missing genes in complex regions. These results suggest that further evaluation is needed on additional species. In this project, we aim to extend this previous work by analyzing new plant genomes. Helixer was applied to the genomes of *Zea mays*² and *Arabidopsis lyrata*⁴. These two species present different levels of genomic complexity. *Zea mays* has a large and highly repetitive genome, while *Arabidopsis lyrata* is closer to a model organism but still less well annotated than *Arabidopsis thaliana*. The predictions generated by Helixer were compared to expert-curated annotations in order to evaluate their accuracy. The comparison focused on several aspects, including the number of predicted genes, the structure of genes, and the agreement between predicted and reference annotations.

Another important aspect of this project is the analysis of annotation versions. A new version of the *Arabidopsis thaliana* annotation (TAIR12)⁸, has recently been released (Public database release: February 2026). The previous one, ARAPORT11⁹ (Official release : June 2017), was used in the previous GBOT analysis. Although ARAPORT11 represented a major improvement in the annotation of the *Arabidopsis thaliana* Col-0 genome, it remains based on older genome assemblies and automated annotation pipelines relying mainly on short-read RNA-seq data. Since its release, advances in sequencing and annotation technologies have enabled more accurate and complete genome references. In this context, TAIR12 was developed using improved long-read genome assemblies of the reference ecotype Col-0, including newly resolved nucleolar organizer regions (NORs) located at the tips of chromosomes 2 and 4¹⁰, leading to a more complete representation of the genome. In addition to refining protein-coding gene models, TAIR12 improves the representation and annotation of complex genomic regions, including TEs and regions that were not previously annotated in ARAPORT11 : centromeric regions, telomeric regions and satellite DNA (**Table 1**). The availability of these two versions allows the evaluation of how improvements in reference annotations affect the assessment of the Helixer tool. Comparing Helixer predictions with ARAPORT11 and TAIR12 may also reveal annotation inconsistencies and regions with significantly modified gene models.

Helixer was applied to *Zea mays* and *Arabidopsis lyrata*, and its predictions were evaluated against expert-curated reference annotations at both exon and coding sequence (CDS) levels using a previously developed comparison framework. Gene structure characteristics, including gene counts, gene lengths, and exon organization, were analyzed. To assess the influence of reference annotation variability, *Arabidopsis thaliana* is analyzed using both TAIR12 and ARAPORT11 annotation versions. Genes shared between both references are used to evaluate structural consistency between annotation sets, and Helixer predictions are assessed with respect to both references. In addition, Helixer-specific predictions in *Zea Mays* (not overlapping with the reference annotation) are analyzed in relation to transposable element-rich regions in order to assess their association with repetitive genomic sequences.

Feature type	Araport11	TAIR12
Protein-coding genes	27,562	26,867
mRNAs	48,265	36,573
Pseudogenes (transcribed/other)	926/3917	2,140
Repeat regions (# of repeats)		
centromeric_repeat	not annotated	65,718
satellite_DNA	not annotated	11,123
telomeric_repeat	not annotated	4,218

Retrotransposons		
Copia_LTR_retrotransposon	1781	1,449
Gypsy_LTR_retrotransposon	4181	3,550
L1_LINE_retrotransposon	1366	876
LINE_element	81	419
LTR_retrotransposon	not annotated	951
SADHU retrotransposon	8	not annotated
SINE_element	131	389
tRNA_SINE_retrotransposon	not annotated	202
Transposons		
CACTA_TIR_transposon (DNA/En-Spm)	941	704
DNA_transposon	1829	772
hAT_TIR_transposon (DNA/HAT)	1035	841
helitron (RC/helitron)	12945	3,666
Mutator_TIR_transposon (DNA/MuDR)	5410	2,375
PIF_Harbinger_TIR_transposon (DNA/Harbinger)	379	379
Tc1_Mariner_TIR_transposon (DNA/Mariner, DNA/Tc1, DNA/Pogo)	590	648

Table 1: Comparison of annotated entities in Araport11 and TAIR12. [11](#)

The green background highlights values that decreased in the updated version, while the gray background indicates values that increased or were newly introduced.

MATERIALS AND METHODS

1. Datasets

The data used in this project were retrieved from the GBOT database, which integrates genomic datasets from multiple organisms. The study focuses on three plant species: *Zea mays* B73 (NAM-5.05), *Arabidopsis lyrata* (Araly v1.0.12), and *Arabidopsis thaliana*, for which both ARAPORT11 and TAIR12 annotation versions were analyzed. For each species, genome assemblies and expert-curated reference annotations were obtained in FASTA and GFF formats. Reference annotations for *Zea mays*, *Arabidopsis lyrata*, and ARAPORT11 were retrieved from the Phytozome database¹, while TAIR12 annotations were obtained from the TAIR resource.

For *Zea mays*, the B73 reference genome (NAM 5.05) and corresponding transposable element annotations were retrieved from the MaizeGDB¹² repository. The genome sequence (FASTA format) and TE annotation file (GFF3 format, EDTA-based annotation) were used for downstream analyses. In addition, the UniProtKB/Swiss-Prot¹³ protein database was used as a curated reference for sequence similarity searches. A local BLAST¹⁴ database was constructed from this resource using standard BLAST formatting tools to enable protein sequence similarity searches.

2. Gene prediction with Helixer

Helixer, a deep learning-based gene prediction tool for eukaryotic genomes, was used to predict gene structures directly from genomic DNA sequences. The model is based on convolutional neural networks trained to infer gene architecture from raw nucleotide sequences without requiring external evidence such as RNA-seq alignments or gene homology information. In its foundational development, Helixer was trained on a dataset comprising 252 eukaryotic genomes, including 192 animal genomes and 60 plant genomes. The animal subset was primarily derived from EnsemblMetazoa (release 45), while plant genomes were obtained from Phytozome (release 13)¹. These datasets include high-quality curated gene annotations in GFF/GFF3 format, integrating information from transcript-supported and expert-curated gene models.

The model learns base-wise probabilistic representations of genomic sequence features, distinguishing functional states such as intergenic regions, exons, introns, untranslated regions, and coding sequences. These probabilities are then decoded into structured gene models defining exon–intron boundaries and coding sequence (CDS) annotations.

For each species (*Zea mays*, *Arabidopsis lyrata*, and *Arabidopsis thaliana*), Helixer was applied to the corresponding genome assemblies in FASTA format. A consistent model configuration was used across all species to ensure comparability across genomes of different sizes and levels of complexity. However for *Zea Mays*, big computational resources were

needed to use Helixer, given its genome size. Therefore, Helixer was runned on usegalaxy.fr¹⁵, a web platform providing access to bioinformatics tools and computational resources. The resulting predictions consist of genome-wide gene models provided in GFF format, including gene coordinates, transcript structures, and CDS regions. These outputs were subsequently parsed and standardized to ensure compatibility with the GBOT comparison framework.

3. Comparison of annotations within GBOT : HelixFlor

GBOT (Genome Browser and Orthologs Tracker) is a tool for the exploration of gene and ortholog relationships, enabling visualization of homologous gene families and ortholog prediction. With this database, the use of SQL queries along with Python code were done to extract and compare data with the comparison tool (**Figure 2**). Moreover, the visualization allows the manual check of the comparison results, and confirms whether the comparison tool performed well on both annotations.

To evaluate Helixer predictions, gene models were compared with expert-curated reference annotations at both exon and coding sequence levels. For this purpose, a dedicated comparison framework named HelixFlor was developed during a previous study conducted within the GBOT project. This tool is implemented using Python, R, and SQL and is currently under active development. It will be released on [the HelixFlor GitHub repository](#).

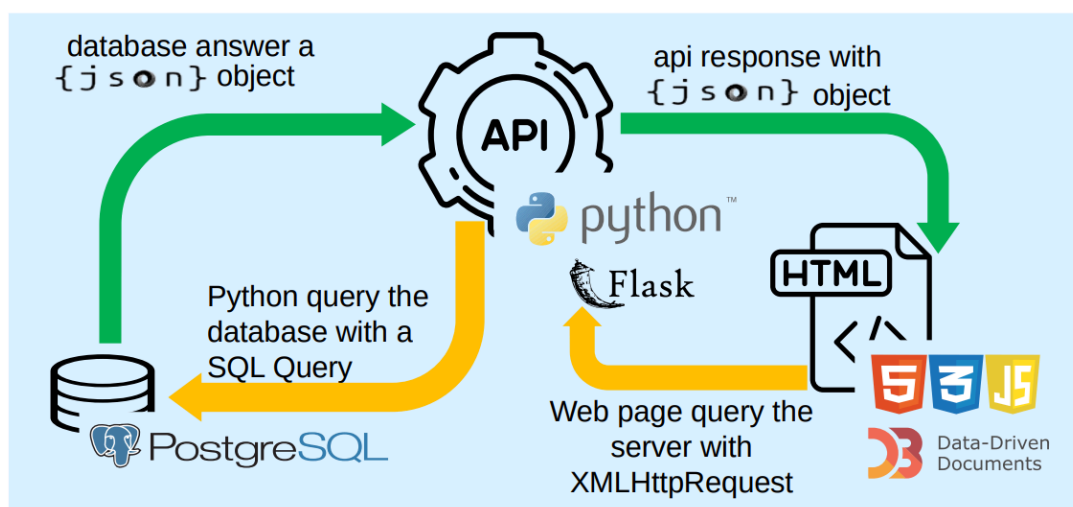


Figure 2: Functional diagram of GBOT.

The web interface uses JavaScript exclusively for the presentation layer, connected to a PostgreSQL database that enables the storage and querying of genomic data. A layer coded in Python acts as an intermediary between the client's browser and the database server via an API. The web interface, which allows the visualisation of homology relationships within gene families across multiple species, is based on WASM and displays database queries via .json outputs.

3.1. Detection and classification of annotation differences

For each pair of features (reference vs Helixer), the HelixFlor tool first verifies strand consistency and genomic orientation. Exon coordinates are then compared using pairwise overlap detection (**Figure 3**), allowing the computation of exon-level similarity scores based on overlapping genomic segments. A global identity score is derived from the mean similarity across matched exons, while missing exons are penalized with zero contribution (**Figure 4**). In parallel, intron structures are reconstructed from exon coordinates and compared to assess conservation of gene architecture between annotations. Genes (loci) were classified as “Added”, “Missed”, or “Common” based on overlap between Helixer predictions and reference transcripts within each locus.

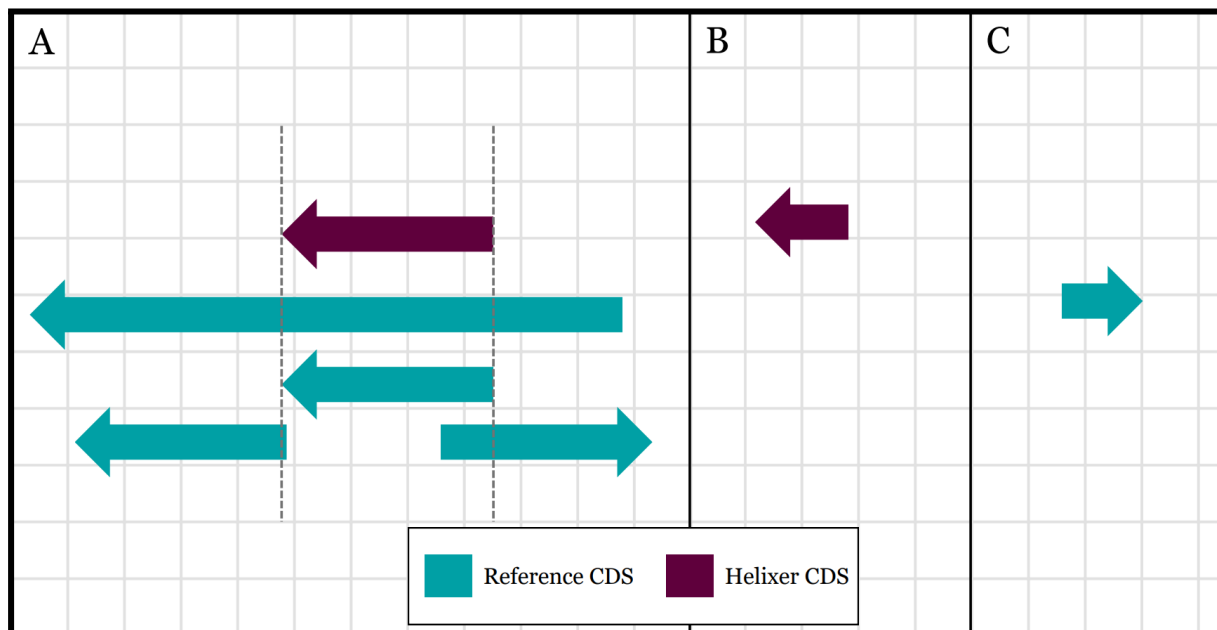


Figure 3: Detection of overlaps between Helixer CDSs and Expert CDSs.

Feature overlaps were determined from genomic coordinates using $\min(\text{stop}_1, \text{stop}_2) - \max(\text{start}_1, \text{start}_2) > 0$ (**A**), meaning even 1 overlapping nucleotide is enough to consider that Helixer annotated an expert reference feature. Non-overlapping features were classified as either added (**B**) or missed by Helixer (**C**). This procedure was used to identify annotation differences between Helixer and the expert reference. **Supplementary Figure 1** provides additional details on how overlapping features of identical or similar lengths were handled.

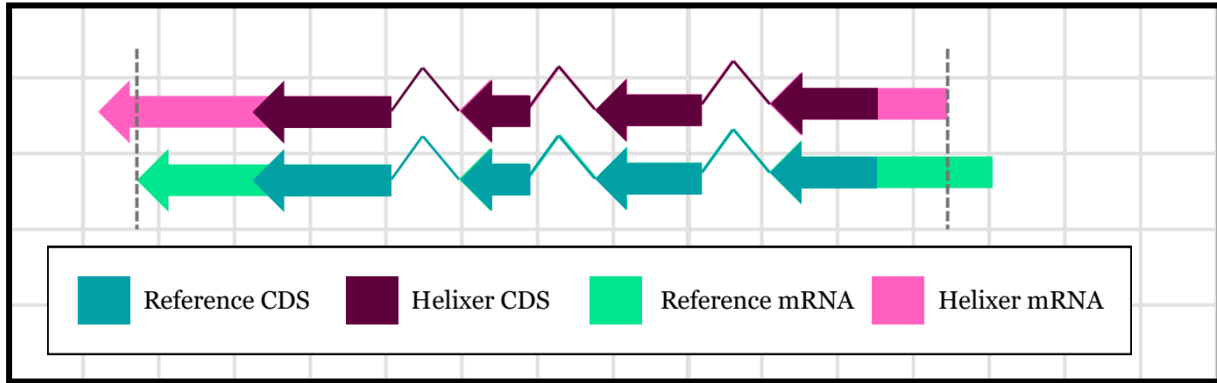


Figure 4: Differences in mRNA annotations in the UTR regions.

Although the CDSs here are identical, this is not the case for the mRNAs, which are almost similar. Helixer exhibits an extension in the 3'UTR compared to the reference, and a deletion in the 5'UTR compared to the reference. The global identity score for this case would be 90%.

Based on exon and intron agreement, each gene is assigned to one of several structural categories, including full matches, partial matches, contained models, fragmented predictions, or non-overlapping predictions. This classification enables the identification of correctly predicted genes, missing genes, added genes, and structural inconsistencies in gene models (**Figure 5**). Rather than using a global transcript comparison tool such as GFFCompare¹⁶, we performed a CDS-constrained analysis. Only mRNAs with identical or nearly identical CDS were retained to control for protein-level conservation.

We then analyzed exon–intron structure and 5'/3' UTR variation within this subset. HelixFlor also handles alternative isoforms by retaining the best-matching transcript based on the highest identity score while storing alternative isoforms for downstream analysis. This prevents redundancy in cases where multiple transcript models overlap the same genomic region. The framework generates structured output tables and BED files for Helixer-specific predictions, reference-specific genes, and shared gene models, enabling downstream visualization and manual inspection.

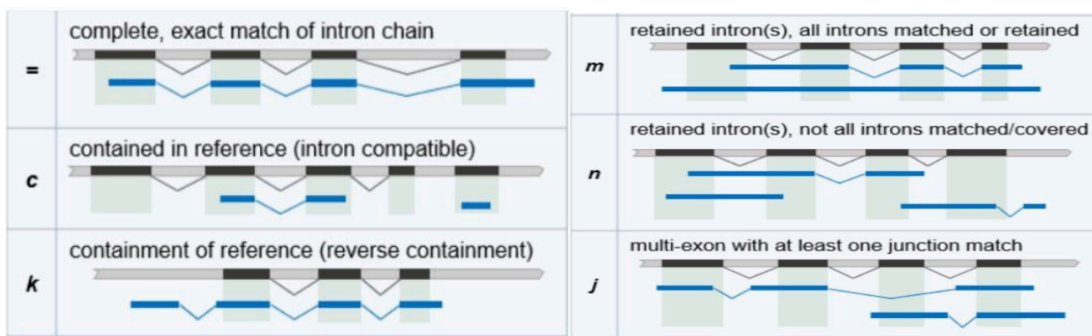


Figure 5: Different types of feature equivalence (established by the E2annot project)

There are several types of feature overlaps, and we drew on the work carried out by the e2annot project (Extend Eukaryote Annotation with transcriptome data) to ensure the tool returns the type of equality for our features.

"=": All exons and introns are exactly the same; "~": All introns are the same but not necessarily the exons; "c": Helixer introns are contained within the reference annotation; "k": Helixer introns contain the reference annotation; "m": All reference introns are covered/matched; "j": At least one match between exons; "n": Not all reference introns are matched. The "~" category was introduced in the HelixFlor project. Transcript pairs were assigned to the "=" category only when their exon–intron structures matched exactly. Any difference in intron boundaries resulted in assignment to the "~" category.

3.2. Functional and sequence-based characterization of predictions

Helixer-specific predictions were further analyzed to assess their potential biological relevance. To provide additional evidence of functionality, Helixer-specific predictions were intersected with Pfam annotations available in the GBOT database. The presence of a Pfam domain was considered supporting evidence for the biological relevance of a predicted gene. Moreover, protein sequences translated from predicted coding sequences (CDS) were compared against the UniProtKB/Swiss-Prot database using BLASTP. Sequence similarity to curated proteins was used as supporting evidence for the validity of these predictions. Orthology relationships were additionally investigated using eggNOG¹⁷ when available.

3.3. Analysis of transposable elements (*Zea Mays* focused)

Helixer-specific predictions in *Zea mays* are further analyzed in relation to transposable elements-rich regions. CDS of these predictions are aligned against the MaizeGDB TE database using BLASTN to assess potential similarity with known repetitive elements. The CDS coordinates (in GFF3 format) were overlapped with TE annotations in GFF3 format using bedtools¹⁸. These analyses are used to evaluate whether gene predictions in repeat-rich regions correspond to misannotated transposable elements.

3.4. TAIR12 and ARAPORT11 annotations comparison

For *Arabidopsis thaliana*, the HelixFlor tool was further used to compare ARAPORT11 and TAIR12 annotations. Genes sharing the same locus identifiers between both annotation versions were compared at the structural level, and Helixer predictions were evaluated with respect to both references. For non-identical gene models, pairwise alignments using MAFFT¹⁹ (Multiple Alignment using Fast Fourier Transform) were used to estimate sequence identity.

3.5. UTR and motif-level analysis (developmental feature)

In addition, untranslated regions (UTRs) predicted by Helixer were extracted and compared to reference annotations. When Helixer predicted a 5' UTR absent from the corresponding reference gene model, the sequence could be further analyzed using Plant-PLMview²⁰ to investigate the presence of over-represented regulatory motifs. This functionality is currently under development.

RESULTS

This section first presents a global evaluation of Helixer across eight plant datasets, including the two newly analyzed genomes, *Zea mays* and *Arabidopsis lyrata*. A detailed structural and functional analysis is then performed for each species, including, for *Zea mays*, an examination of Helixer-specific predictions in relation to MaizeGDB transposable element annotations, before comparing Helixer's performance against both the ARAPORT11 and TAIR12 annotations of *Arabidopsis thaliana*.

1. Global overview of Helixer predictions across species

To provide an overall assessment of Helixer predictions, gene models were first classified into three categories relative to reference annotations: common, added, and missing genes. In seven of the eight datasets, more than 70% of genes were classified as common between Helixer and the corresponding reference annotation. The proportion of Helixer-specific predictions varied substantially among species, reaching 23.5% in *Rosa chinensis*, whereas *Arabidopsis thaliana* (ARAPORT11) showed the highest level of agreement with the reference annotation (92.3%) (**Figure 6**). This relatively lower agreement is likely attributable to the early (v1) version of the *Rosa chinensis* reference annotation²¹, combined with the genome's high TE content, both of which can inflate apparent disagreement between Helixer and the reference.

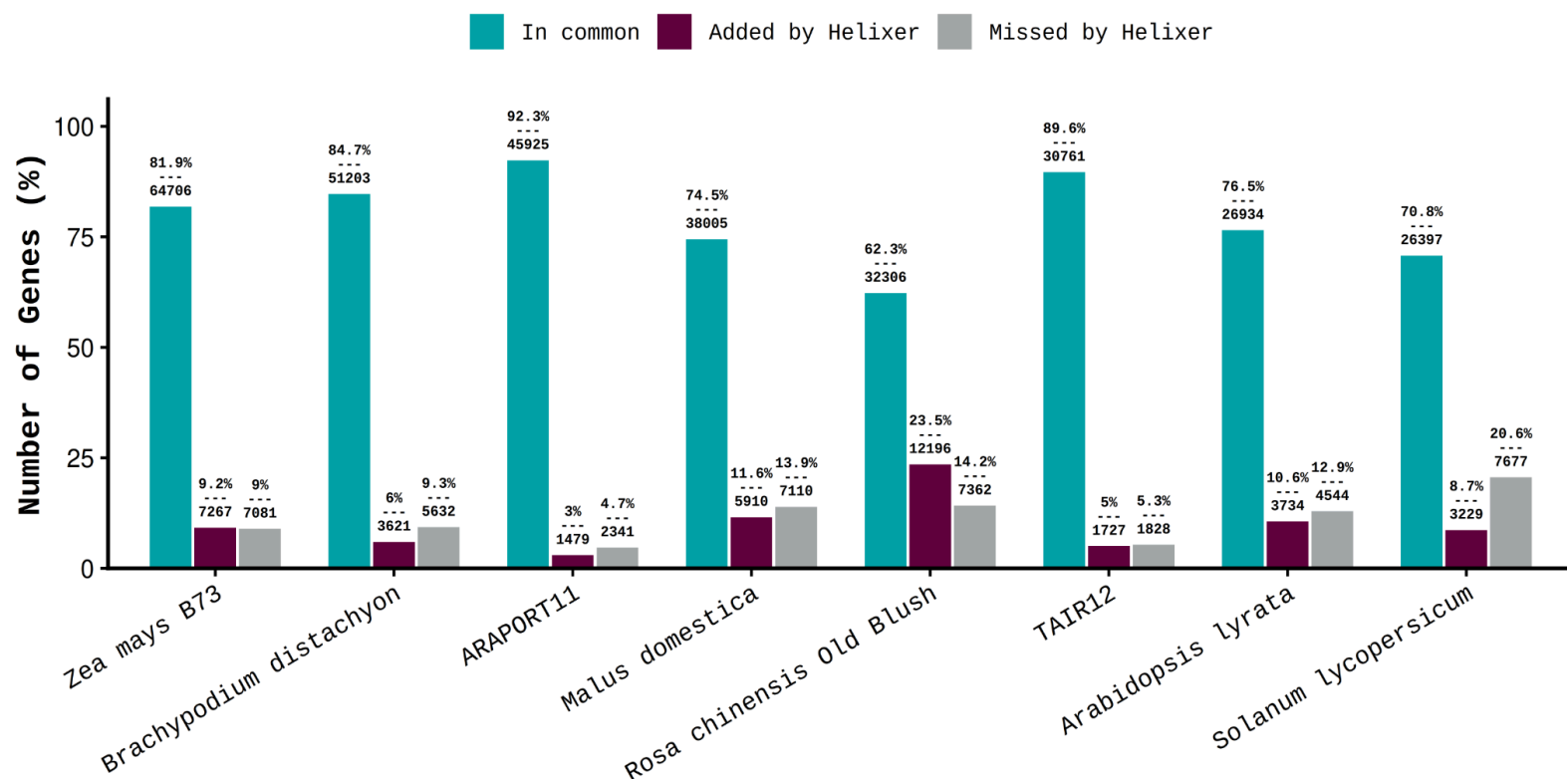


Figure 6: Evaluation of Helixer gene prediction performance across 8 plant species.

Comparison of CDS annotations predicted by Helixer against reference annotations for *Zea mays* B73, *Brachypodium distachyon*, ARAPORT11, *Malus domestica*, *Rosa chinensis* Old Blush, TAIR12, *Arabidopsis lyrata*, and *Solanum lycopersicum*. Bars represent the percentage and number of genes categorized as: "In common" (matching reference), "Added by Helixer" (novel predictions not in the reference), and "Missed by Helixer" (reference genes not predicted by the tool).

Among the analyzed genomes, *Arabidopsis lyrata* was selected for further investigation because, although it is not a model species, it is closely related to the model plant *Arabidopsis thaliana*. This close evolutionary relationship enables the assessment of Helixer performance in a less extensively annotated genome while maintaining a direct point of comparison with a well-characterized model species.

2. Evaluation on *Arabidopsis lyrata*

A detailed evaluation of Helixer predictions was performed on *Arabidopsis lyrata* (*A. Lyrata*). Gene structure agreement was assessed at both exon and coding sequence (CDS) levels, including gene counts, gene lengths, and exon organization. In addition, untranslated region (UTR) structures were analyzed when available, allowing comparison of 5' and 3' UTR extensions between Helixer predictions and reference annotations.

Reference annotations omitted by Helixer ("Missed") were predominantly shorter (median length 413 bp) and contained fewer exons (mean = 2). In contrast, Helixer-specific predictions ("Added") were longer (median length 982 bp) and contained a higher number of exons (mean = 3.4). Functional annotation analyses provided complementary information for these gene sets. In the GBOT Pfam domain scan, no Pfam hits were detected among the added gene models ($n = 0$), whereas 8,477 common models and 969 missed models contained at least one Pfam domain annotation. In contrast, comparison with UniProtKB identified 1,114 added Helixer predictions, out of 3,734 added predictions, showing significant sequence similarity ($\geq 50\%$ identity and $\geq 60\%$ coverage) (**Figure 7**).

An analysis of transcript untranslated regions (UTRs) in *A. lyrata* high-concordance gene models revealed differences between Helixer and reference annotations (**Table 2**). For the 14,184 transcripts with identical CDS structures ($\geq 90\%$ identity), Helixer predicted longer UTRs than the reference genome. Mean 5' UTR length was 241 bp in Helixer compared to 105 bp in the reference, while 3' UTRs averaged 206 bp and 152 bp, respectively. A similar pattern was observed for the 1,161 "Almost Identical" transcripts (85–99% CDS identity), with mean Helixer UTR lengths of 232 bp (5') and 170 bp (3'), compared to 98 bp and 138 bp in the reference. Despite strong conservation at the CDS level, mean sequence identity within UTR regions was 73% for identical models and 70% for near-identical models; most transcript pairs belonged to the "~" category, indicating conserved exon–intron boundaries.

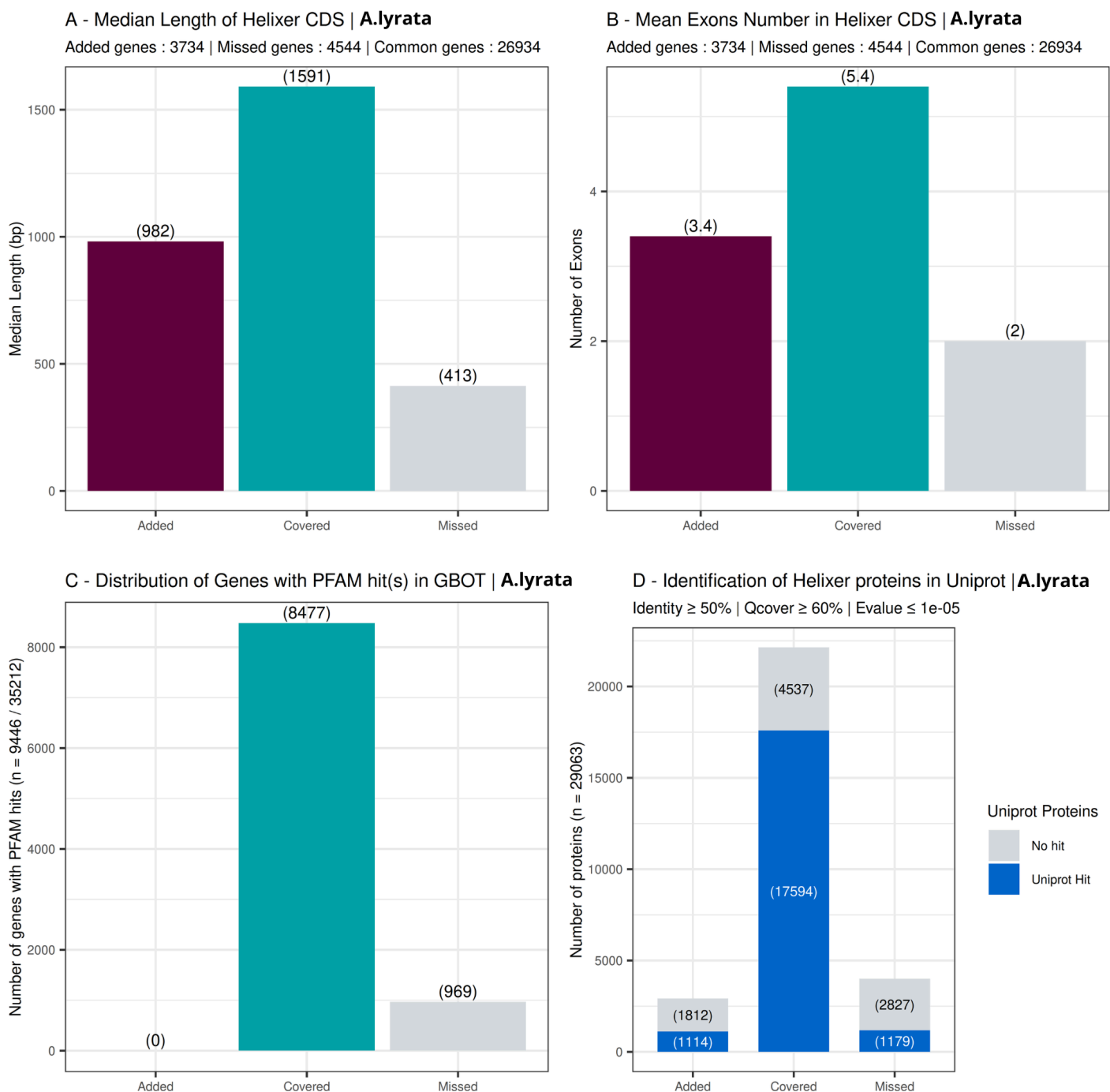


Figure 7: Structural and functional characterization of Helixer-predicted gene categories in *A.lyrata*.

Comparative analysis of genes classified as novel predictions ("Added"), matching reference predictions ("Covered/Common"), and undetected reference genes ("Missed"). Total counts per category are indicated at the top of the panels. **(A)** Median length of Helixer coding sequences (CDS) in base pairs (bp), with exact median values displayed above each bar. **(B)** Mean number of exons per gene across the three annotation categories. **(C)** Distribution of genes containing at least one functional PFAM hit within the GBOT database. **(D)** Validation and identification of Helixer-predicted proteins against the UniProt database. Stacked bars represent proteins with a confirmed sequence match ("UniProt Hit") versus those without ("No Hit") under stringent filtering criteria (Identity $\geq 50\%$, Qcover $\geq 60\%$, E-value $\leq 1e-05$). **Supplementary Figure 2** provides additional details on the distribution of the Helixer-specific proteins with their respective identity percentage in UniprotKB.

Arabidopsis lyrata: Genome annotations summary							
mRNAs Structural analysis for specific CDS structures							
CDS Structure	Number	Mean Identity %	Most Frequent Type	5'UTR Helixer Length	3'UTR Helixer Length	5'UTR Reference Length	3'UTR Reference Length
Identical	14184	73 ¹	~	241 ^{1,2}	206 ^{1,2}	105 ^{1,2}	152 ^{1,2}
Almost Identical	1161	70 ¹	~	232 ^{1,2}	170 ^{1,2}	98 ^{1,2}	138 ^{1,2}

¹ Only mRNAs linked to the CDS above were compared here.

² Average Length of UTR regions

Table 2: Comparative analysis of 5' and 3' UTR structures in high-concordance *A.lyrata* transcripts.

Structural features of untranslated regions (UTRs) were assessed for transcripts classified by coding sequence (CDS) agreement with the reference genome. Transcripts were grouped into two categories based on CDS identity thresholds: "Identical" ($\geq 90\%$ identity; $n = 14184$) and "Almost Identical" (85–99% identity; $n = 1161$). The table reports transcript counts, mean sequence identity (%) for the transcripts, and average 5' and 3' UTR lengths (bp). UTR metrics were computed only for transcripts associated with the corresponding CDS categories, and lengths represent mean nucleotide counts.

Helixer reconstructed conserved gene structures in *A.lyrata* (**Supplementary Figures 3A & 3B**). In addition, "Added" and "Missed" gene models were generally shorter and exhibited simpler exon structures than "Covered" genes. Differences in transcript boundaries and UTR lengths were also observed, with Helixer generally predicting longer UTRs than the reference annotation despite high CDS conservation. A similar pattern was observed against both *A. thaliana* annotations, with "Covered" genes consistently longer and more exon-rich than "Added" or "Missed" models for ARAPORT11 (**Supplementary Figures 6A & 6B**) and TAIR12 (**Figures 12A & 12B**). Against TAIR12, Helixer 5' UTRs were longer than the reference on average (241 bp vs. 203 bp), whereas 3' UTRs were shorter (206 bp vs. 229 bp) (**Table 5**). A comparable trend was observed for ARAPORT11, with Helixer 5' UTRs averaging 241 bp versus 255 bp in the reference, and 3' UTRs averaging 207 bp versus 276 bp (**Supplementary Table 3**).

3. *Zea mays* Results structure

Overall, Helixer performance in *Zea mays* shows a similar pattern to that observed in *Arabidopsis lyrata* (**Supplementary Figures 4 & 5**), with comparable distributions of common, added, and missed gene models. However, differences are observed at the level of untranslated regions (UTRs). Helixer 5' UTRs are consistently longer than reference annotations by approximately 43 bp in both identical and almost-identical transcript categories. In contrast, 3' UTRs are shorter in Helixer predictions, with mean differences of approximately 137 bp in the identical category and 76 bp in the almost-identical category. In addition, mean sequence identity in *Zea mays* is approximately 10% higher than in *A. lyrata* (~80% vs ~70%) (**Supplementary Table 1**).

Given that complex plant genomes like maize are highly repetitive, we hypothesized that non-coding or silenced transposable elements (TEs) might be driving the over-prediction of the Helixer-specific unannotated, unhit proteins. Spatial coordinates of the Helixer-specific models were intersected with MaizeGDB TE annotations, followed by rigorous BLASTn confirmation. A total of 1,949 genes out of 7,267 Helixer-specific showed high-confidence genomic overlap with annotated TEs, supported by BLASTn validation exhibiting strong sequence similarity (E-values ranging from 0 to 1×10^{-46}), high query coverage (90–100%), and nucleotide identity between 83.2% and 100% (**Table 3**).

BLASTn Summary Statistics	
Global ranges across all Helixer-specific predictions having the same genomic positions as MaizeGDB TEs	
Metric	Value
Number of genes	1949
E-value range	0 - $1e-46$
QCover range	90 - 100%
Identity range	83.2 - 100%

Table 3: BLASTn alignment statistics for Helixer-specific predictions overlapping MaizeGDB-annotated transposable elements.

BLASTn alignment statistics for the 1,949 Helixer-specific predictions overlapping MaizeGDB-annotated transposable elements, showing high-confidence matches characterized by query coverage of 90–100% and sequence identity of 83.2–100%, indicative of strong correspondence with annotated transposable element sequences.

To assess whether TE overlap contributes to the absence of reference-based functional annotation, TE intersections were stratified according to UniProt mapping success (**Figure 8**). Among predictions lacking a UniProt match (“No hit”), 27% (938/3,457) overlapped annotated TEs. In contrast, TE overlap was slightly lower in the sequence-supported subset

(“UniProt Hit”), with approximately 25% (190/753; identity >70%, query cover > 60% and Evalue < 1×10^{-05}) showing intersection with TE regions. However, a chi-square test indicated that this association was not statistically significant ($X^2 = 1.0443$, $df = 1$, $p = 0.3068$), suggesting no strong dependence between TE overlap and lack of protein-level annotation under the applied criteria.

These results highlight the impact of transposable element-rich regions on the characterization of Helixer-specific predictions in maize.

Association between TEs overlap and absence of UniProt protein hits

QCover filtered > 60% | Evalue < $1e-05$ | Identity filtered >70% | Total proteins: 7267 | Hits: 753 | TEs: 1128/1949

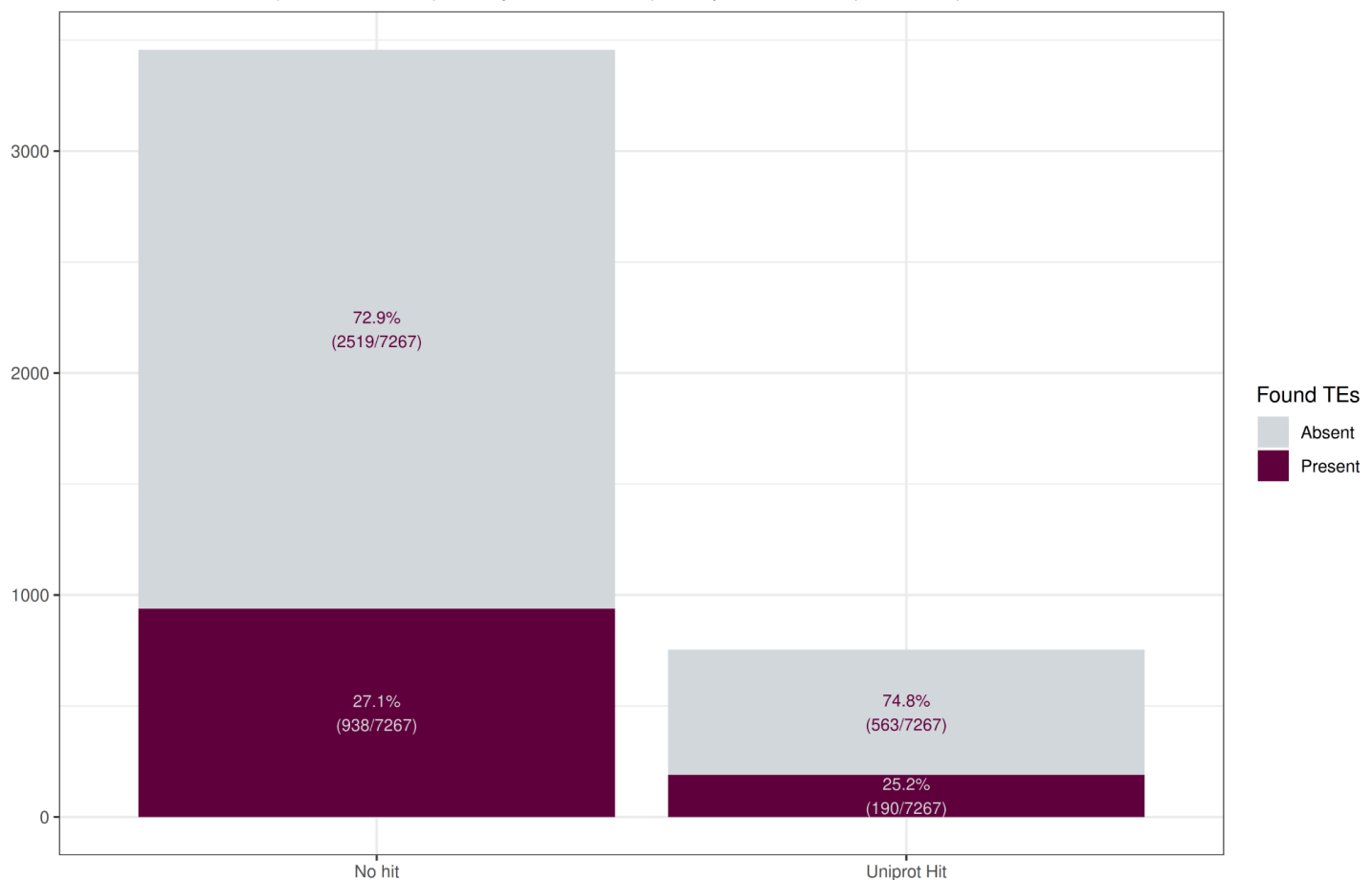


Figure 8: Functional and transposable element stratification of Helixer novel gene predictions in *Zea mays*

Stacked bar chart showing the 7,267 novel gene models predicted by Helixer in the *Zea mays* B73 genome, stratified by UniProt alignment status. Predictions are divided into models lacking sequence similarity to UniProt (“No hit”, $n = 3,457$) and those with significant database matches (“UniProt Hit”, $n = 753$). Each category is further subdivided based on genomic overlap with annotated transposable elements, distinguishing TE-overlapping (“Present”) from non-overlapping (“Absent”) predictions. Percentages are calculated relative to the total set of novel predictions ($n = 7,267$). Among TE-overlapping predictions ($n = 1,949$), 1,128 fall within the UniProt-stratified categories under the applied filtering criteria.

4. Comparison of TAIR12 and ARAPORT11 annotations in *A.thaliana*

To evaluate how alternative reference annotations may influence Helixer results, gene models were compared between the TAIR12 and ARAPORT11 annotations (**Figure 9**). Although a large set of gene models is shared between both releases, differences remain in annotation-specific content, with ARAPORT11 containing considerably more unique models than TAIR12. This asymmetry reflects differences in transcript representation and annotation scope between the two releases. Moreover differences in mean CDS length and exon number further illustrate structural divergence between the datasets (**Table 4**).

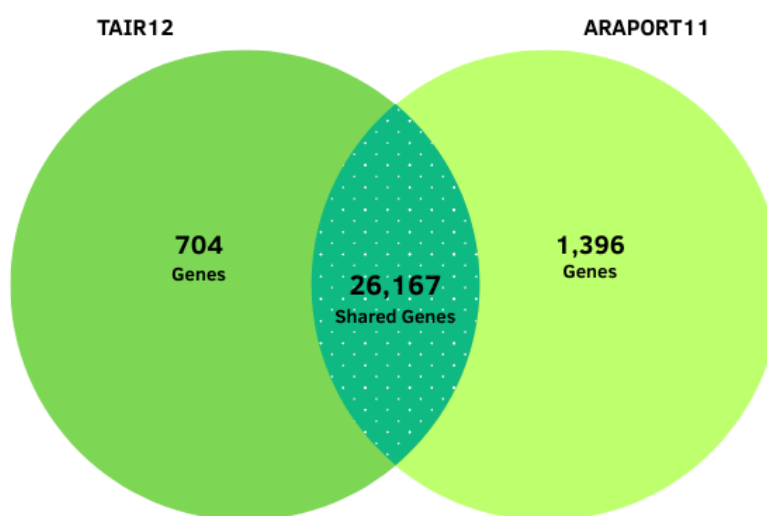


Figure 9: Overlap of annotated gene models sharing common identifiers between the TAIR12 and ARAPORT11 reference annotations of *A.thaliana*.

Venn diagram showing the distribution of shared and annotation-specific gene models between the TAIR12 and ARAPORT11 reference annotations of *Arabidopsis thaliana*. A total of 26,167 gene models are shared between both annotations, while 704 and 1,396 models are unique to TAIR12 and ARAPORT11, respectively.

TAIR12: Genome annotations summary		
TAIR12 VS ARAPORT11 : CDS		
Number	Length Difference (bp)	Exons Number Difference
30443	76.2486	0.2

Table 4: Coding sequence length and exon count differences between TAIR12 and ARAPORT11 shared gene models

Structural comparison of the 30,443 CDS models sharing identical identifiers between the TAIR12 and ARAPORT11 annotations. Unlike **Figure 9**, which reports gene-level overlap, this comparison is performed at the transcript level and therefore includes both primary and alternative transcript isoforms explaining the higher number of shared CDS models. The table reports the mean absolute difference in CDS length (bp) and exon count between corresponding CDS models, calculated across all shared models.

Protein sequences were compared across the 30,443 shared loci common to TAIR12 and ARAPORT11 (**Supplementary Table 2**). The majority of shared loci exhibit complete sequence conservation, with 74.2% ($n = 22,591$) producing identical amino acid sequences across annotations. However, 25.8% ($n = 7,852$) encode divergent protein sequences despite sharing identical gene identifiers. This indicates that a substantial fraction of shared loci differs at the protein sequence level between the two reference annotations.

Global multiple sequence alignments using MAFFT were performed on the 7,852 divergent protein pairs identified between TAIR12 and ARAPORT11 (**Figure 10**). The resulting distribution shows that most sequence differences remain within high similarity ranges. Out of these divergent protein pairs, a total of 77.3% ($n = 6,070$) of models exhibit pairwise identity above 90%. An additional 16.4% ($n = 1,288$) fall within the 70–90% identity range, while only 3.8% ($n = 298$) and 2.5% ($n = 196$) show identities in the 50–70% and <50% ranges, respectively. Overall, the majority of divergent loci retain high sequence similarity despite differences between annotation releases.

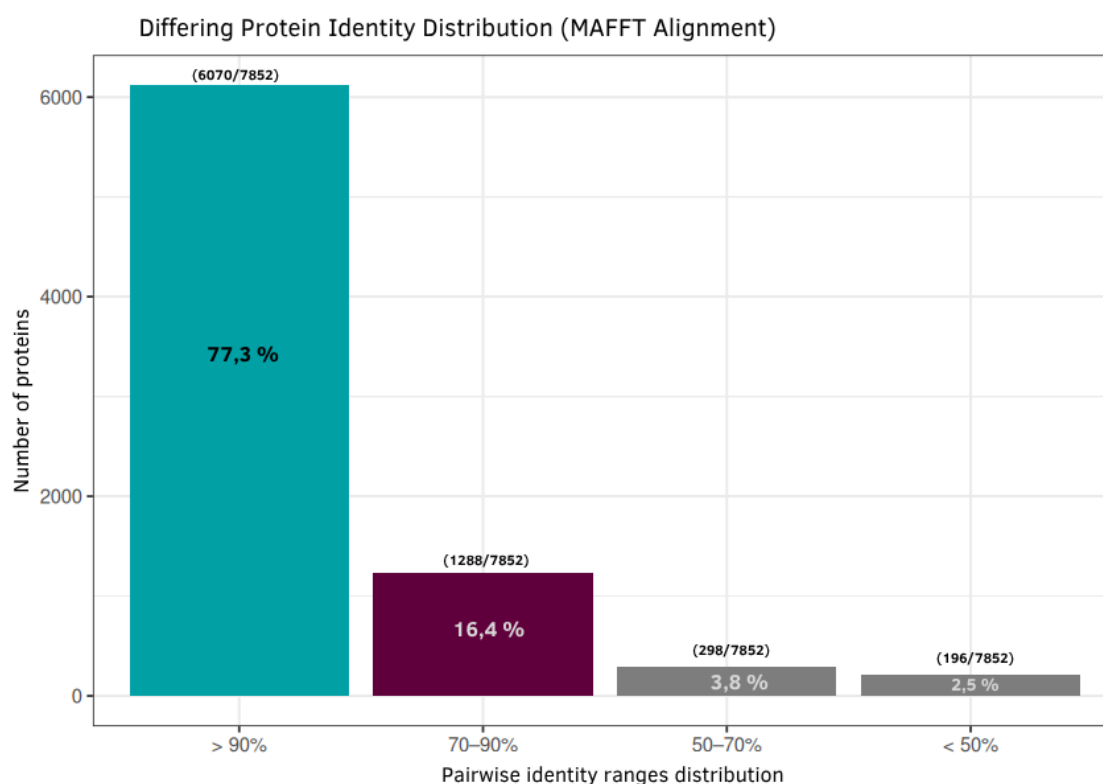


Figure 10: Pairwise sequence identity distribution of divergent reference protein models following MAFFT global alignment.

Distribution of pairwise amino acid sequence identity across 7,852 divergent protein models identified between TAIR12 and ARAPORT11. Sequence identities were obtained from global MAFFT alignments and grouped into conservation bins (>90%, 70–90%, 50–70%, and <50%). Bars indicate the proportion and number of protein pairs in each identity range relative to the total divergent set.

To quantify how Helixer predictions align with established reference annotations at the transcript level, absolute CDS length differences were computed for gene models shared between ARAPORT11 and TAIR12 (**Figure 11**). These CDS length differences were used to assess structural deviation between Helixer predictions and the reference annotations. The resulting distribution is centered around a median difference of approximately 200–300 bp across both comparisons. This indicates a consistent pattern of CDS length deviation between Helixer predictions and both reference datasets.

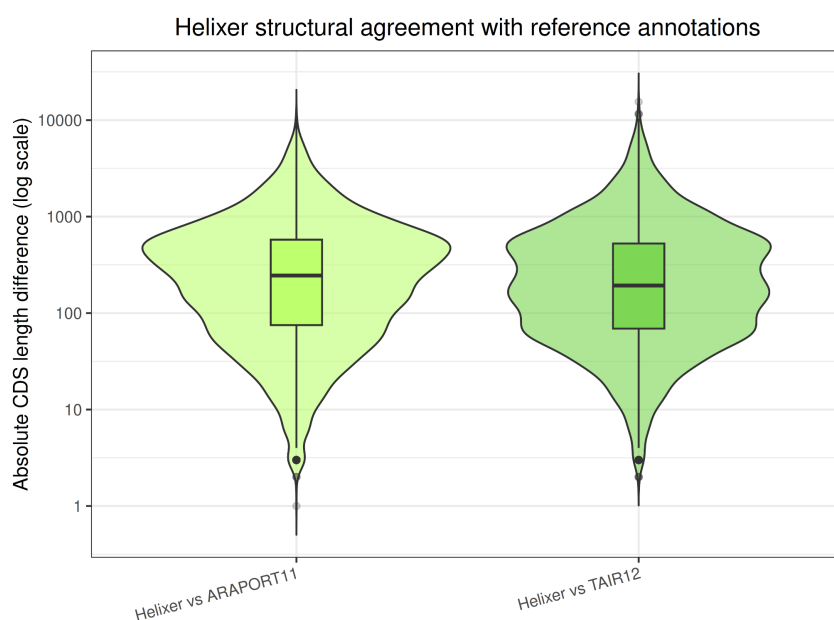


Figure 11: Distribution of CDS length differences between Helixer predictions and reference annotations.

Violin and box plots showing the distribution of absolute CDS length differences (log-scaled base pairs) between Helixer models and matching reference gene models. Comparisons are shown separately for ARAPORT11 ($n = 1,376$ divergent CDS) and TAIR12 ($n = 3,229$ divergent CDS). The median is indicated by the central line, and box limits represent the interquartile range.

5. Comparison of TAIR12 and Helixer annotations

Comparing Helixer predictions to the TAIR12 reference annotation, the overall pattern was consistent with that observed in *A. lyrata*: “Covered” genes were longer and contained more exons, whereas “Added” and “Missed” genes were generally shorter (**Figure 12**). Focusing on the 1,727 Added proteins, 40.6% showed a significant UniProt match (QCover > 60%). The distribution of sequence identities included both near-perfect matches and lower-identity alignments (**Figure 13A**). PFAM domain annotations were more frequent among UniProt-matching proteins (77%) than among proteins without a UniProt hit (29%) (**Figure 13B**). Functional annotation was further assessed using EggNOG on a subset of 904 proteins lacking a significant UniProt match. Functional assignments were recovered for 61.3% of these proteins, whereas 38.7% remained unannotated (**Figure 13C**). Similar trends were observed in ARAPORT11. (**Supplementary Figures 6 & 7**)

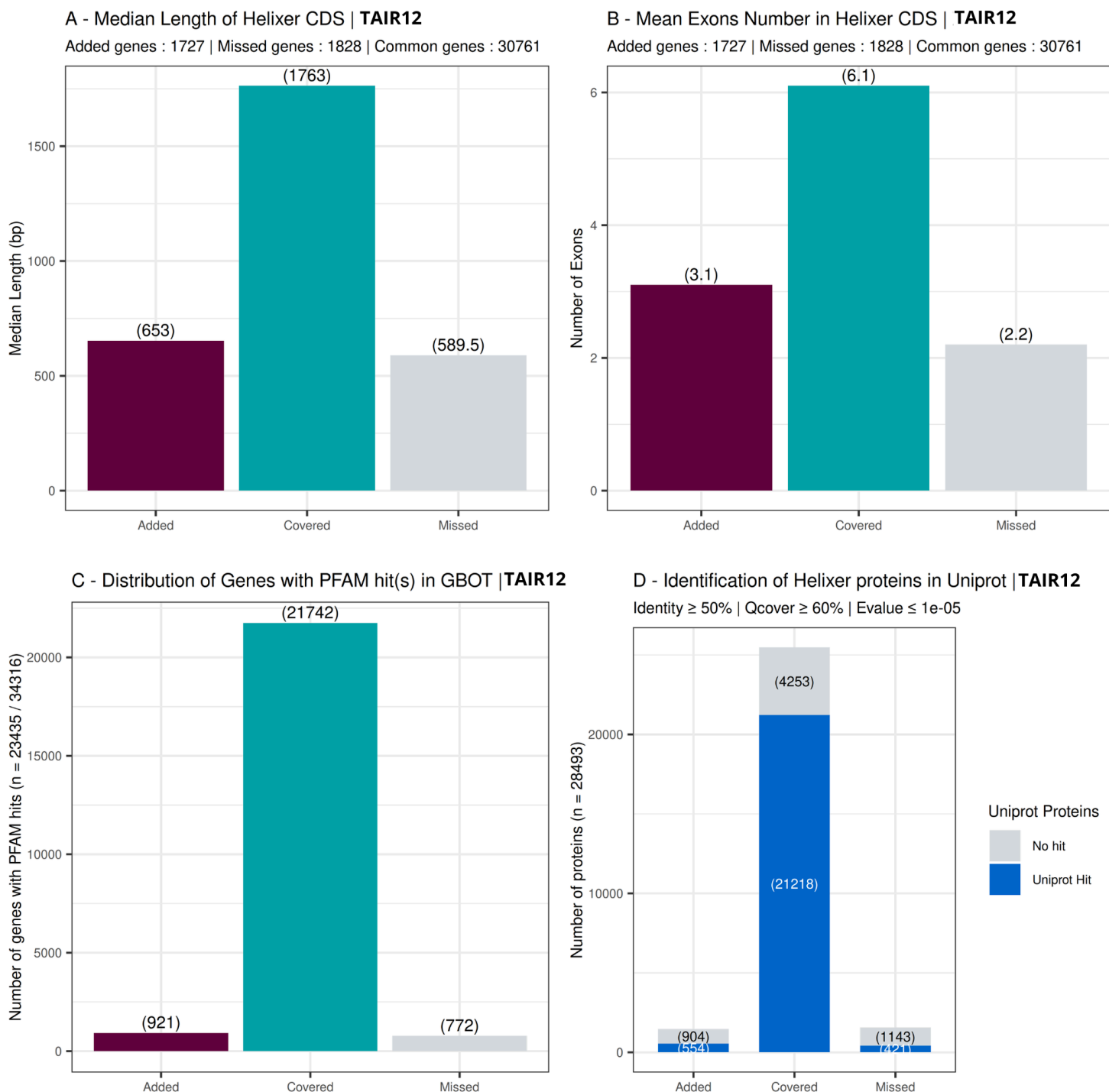
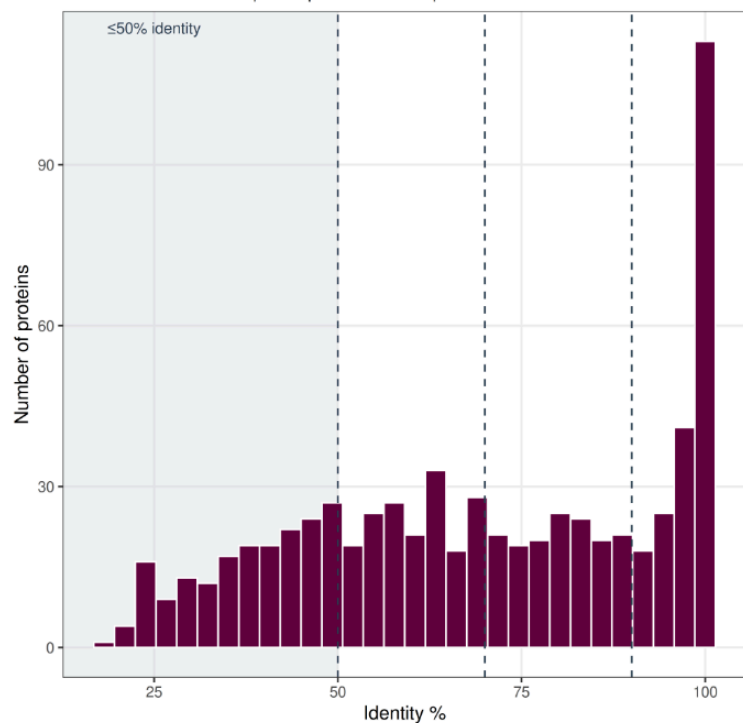


Figure 12: Structural and functional characterization of Helixer-predicted gene categories in TAIR12.

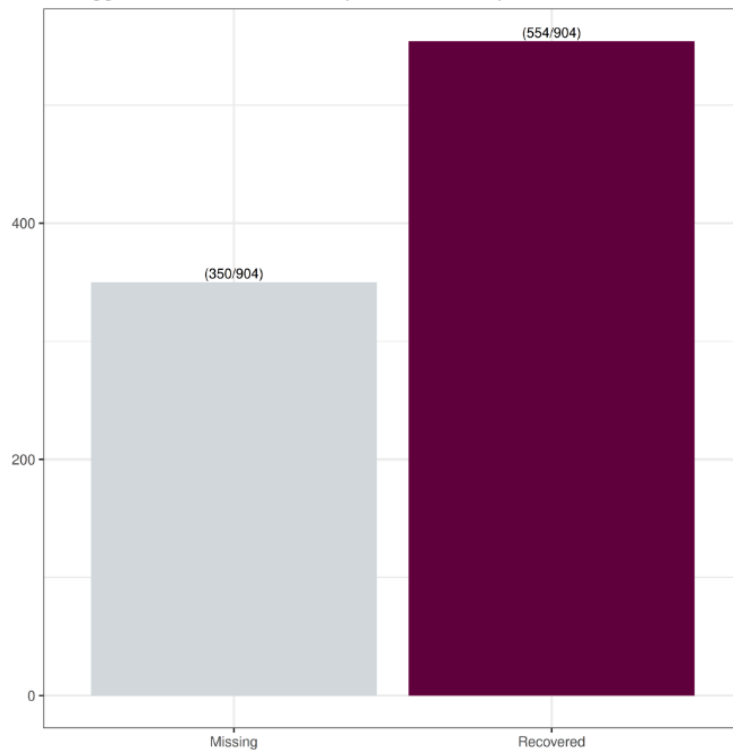
Comparative analysis of genes classified as novel predictions ("Added", $n = 1,727$), matching reference predictions ("Covered/Common", $n = 30,761$), and undetected reference genes ("Missed", $n = 1,828$) relative to the TAIR12 annotation. Total counts per category are indicated at the top of the panels. **(A)** Median length of Helixer coding sequences (CDS) in base pairs (bp), with exact median values displayed above each bar. **(B)** Mean number of exons per gene across the three annotation categories. **(C)** Distribution of genes containing at least one functional PFAM hit within the GBOT database. **(D)** Validation and identification of Helixer-predicted proteins against the UniProt database. Stacked bars represent proteins with a confirmed sequence match ("UniProt Hit") versus those without ("No Hit") under stringent filtering criteria (Identity $\geq$ 50%, Qcover $\geq$ 60%, E-value $\leq$ 1e-05).

A - Identity distribution of Helixer proteins in Uniprot

QCover filtered > 60% | Total proteins: 1727 | Hits: 701



C - EggNOG annotations for Uniprot unmatched proteins



B - PFAM support across Helixer-specific proteins and UniProt mappings

QCover filtered > 60% | Total proteins: 1727 | Hits: 701

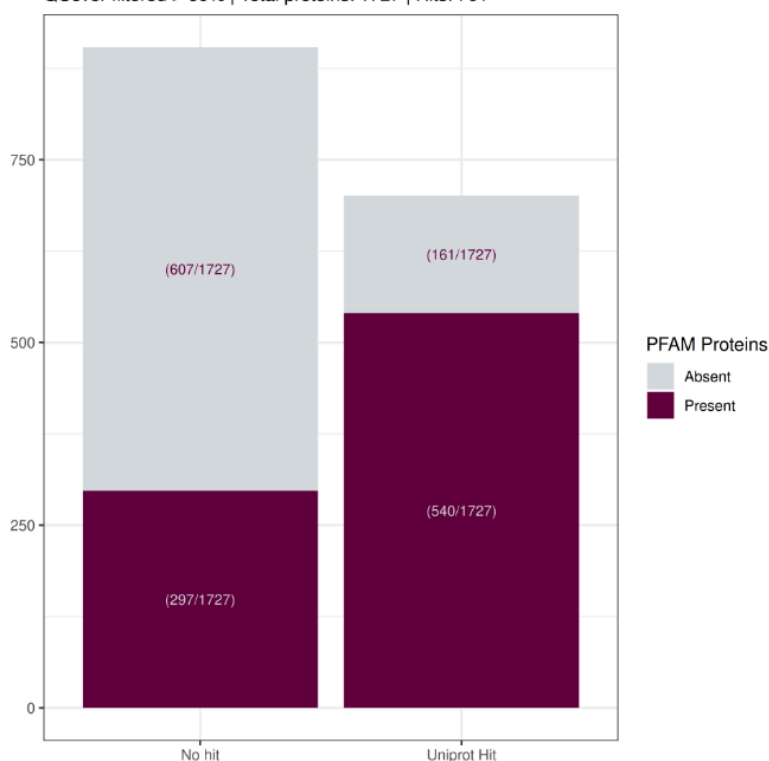


Figure 13: UniProt and PFAM/EggNOG functional support for Helixer-specific (Added) proteins relative to TAIR12.

(A) Distribution of UniProt identity percentages for Added proteins with a significant hit (query coverage > 60%; total proteins = 1,727, hits = 701). Dashed lines mark 50%, 70%, and 90% identity thresholds. **(B)** Number of Added proteins with and without a detectable PFAM domain, stratified by UniProt hit status. **(C)** Number of UniProt-unmatched Added proteins ($n = 904$) for which functional annotation was recovered via EggNOG ("Recovered") versus those that remained unannotated ("Missing").

A comparable UTR-level analysis was performed for high-concordance gene models shared between Helixer and TAIR12 (**Table 5**). The 20,142 identical CDS transcripts showed a mean mRNA sequence identity of 90%, with Helixer predicting longer 5' UTRs (241 bp vs 203 bp in TAIR12) and shorter 3' UTRs (206 bp vs 229 bp). The 568 "Almost Identical" transcripts showed a similar pattern, with Helixer 5' and 3' UTR lengths of 223 bp and 148 bp compared to 160 bp and 224 bp in TAIR12. A similar trend was observed with ARAPORT11, although reference UTR lengths differed between annotation versions (**Supplementary Table 3**).

TAIR12: Genome annotations summary							
mRNAs Structural analysis for specific CDS structures							
CDS Structure	Number	Mean Identity %	Most Frequent Type	5'UTR Helixer Length	3'UTR Helixer Length	5'UTR Reference Length	3'UTR Reference Length
Identical	20142	90 ¹	~	241 ^{1,2}	206 ^{1,2}	203 ^{1,2}	229 ^{1,2}
Almost Identical	568	88 ¹	~	223 ^{1,2}	148 ^{1,2}	160 ^{1,2}	224 ^{1,2}

¹ Only mRNAs linked to the CDS above were compared here.

² Average Length of UTR regions

Table 5: Comparative analysis of 5' and 3' UTR structures in high-concordance TAIR12 transcripts.

Structural features of untranslated regions (UTRs) were assessed for transcripts classified by coding sequence (CDS) agreement between Helixer predictions and the TAIR12 reference annotation. Transcripts were grouped into two categories based on CDS identity thresholds: "Identical" ($\geq 90\%$ identity; $n = 20,142$) and "Almost Identical" ($85-99\%$ identity; $n = 568$). The table reports transcript counts, mean sequence identity (%) for the transcripts, and average 5' and 3' UTR lengths (bp) for Helixer and for the TAIR12 reference. UTR metrics were computed only for transcripts associated with the corresponding CDS categories, and lengths represent mean nucleotide counts.

Overall, comparisons against ARAPORT11 and TAIR12 produced broadly similar results. Covered genes consistently displayed longer CDS lengths and higher exon counts than "Added" or "Missed" genes, while differences in UTR lengths were observed for both reference annotations. Most variations between the two comparisons corresponded to differences in gene content and transcript annotation between ARAPORT11 and TAIR12.

DISCUSSION

This project aimed to extend a previous GBOT evaluation of Helixer by applying the tool to two additional plant genomes, *Zea mays* and *Arabidopsis lyrata*, while also examining how the choice of reference annotation version influences the assessment of a gene prediction tool, using *Arabidopsis thaliana* as a case study. Each of these objectives is discussed below in light of the results obtained.

Extension of Helixer evaluation to new species. Including *Zea mays* and *A.lyrata* in the GBOT comparison framework allowed Helixer to be assessed across two contrasting genomic contexts: a large, highly repetitive genome and a non-model genome closely related to a well-characterized reference species. The overall structural concordance observed for both species supports the conclusions of the earlier GBOT study, suggesting that Helixer's performance is not restricted to the genomes used in its original benchmarking. This is particularly relevant for *A.lyrata*, which was not part of Helixer's training data: the tool nonetheless reconstructed conserved gene architectures, although the UniProt-supported subset of Helixer-specific predictions was dominated by hits to *A.thaliana* proteins. This pattern highlights the dependence of automated annotation on closely related, well-annotated model species and the increased uncertainty in genomes lacking such references, even when Helixer performs well structurally. Also, Helixer tends to add genes that are shorter and structurally simpler, while a similar trend is observed among missed reference genes. This pattern suggests a bias related to gene architecture, where shorter or less complex exon structures are more difficult to consistently capture, and may also reflect uncertainty in the annotation of such gene models in reference datasets.

Influence of genomic repeat content on prediction accuracy. The TE-association analysis performed in *Zea mays* was motivated by the hypothesis that Helixer-specific, functionally unsupported predictions might correspond to mis-annotated transposable elements rather than genuine coding genes. While a substantial fraction of these predictions did overlap annotated TEs, the lack of a statistically significant association with the absence of UniProt support indicates that TE overlap alone does not fully explain the functional uncertainty of Helixer-specific models. A likely confounding factor is that the TE database used for this analysis includes both active and degraded elements. Restricting this analysis to a curated set of active *Zea mays* TEs would provide a more stringent test of this hypothesis and should be considered a priority refinement of this analysis.

Impact of reference annotation version on tool evaluation. The comparison between TAIR12 and ARAPORT11 was designed to determine whether the conclusions drawn from benchmarking Helixer depend on the maturity of the reference annotation itself. Helixer was trained on Phytozome data, which includes TAIR10, an earlier version of the *A.thaliana* annotation; this training origin likely explains the consistently strong agreement observed between Helixer and both ARAPORT11 and TAIR12, since both annotations are themselves downstream refinements of the same reference genome lineage. Importantly, the fact that Helixer performed comparably well against TAIR12, an annotation released after Helixer's

training and therefore unseen by the model, strengthens the conclusion that this agreement reflects genuine generalization rather than memorization of training data. The presence of CDS-level differences between TAIR12 and ARAPORT11 for shared loci suggests that reference annotations are not entirely consistent. This implies that part of the variation attributed to Helixer in single-reference analyses may originate from differences between annotation versions rather than model errors, reinforcing the value of multi-reference evaluation.

Genome complexity as a driver of prediction difficulty. Across the eight species considered in the global comparison, the proportion of Helixer-specific predictions varied considerably, reaching its highest value in *Rosa chinensis*. As noted in the Results, the elevated proportion of Helixer-specific predictions in *Rosa chinensis* likely reflects both the early (v1) status of its reference annotation and the genome's high TE content²¹, rather than the species absence from Helixer's training set alone. A dedicated investigation of *Rosa chinensis*, applying the same TE-overlap and functional characterization strategy used for maize, would help determine whether genome complexity rather than training set composition is the main driver of this difference.

UTR boundaries as a weaker aspect of prediction. A recurrent observation across *A.lyrata*, *Zea mays*, and the TAIR12/ARAPORT11 comparison was the systematic divergence between Helixer and reference annotations at the level of UTR length, despite strong agreement at the CDS level. This suggests that Helixer's predictive accuracy is not uniform across gene structure components, and that UTR boundaries, which are typically less constrained by CDS conservation and more dependent on transcriptomic evidence, remain a weaker aspect of *ab initio* prediction. Notably, Helixer's predicted UTR lengths were nearly identical whether compared to TAIR12 or ARAPORT11, while sequence identity was consistently higher against TAIR12, and ARAPORT11 reference UTRs were markedly longer than TAIR12's. This indicates that the divergence originates largely from the references themselves rather than from instability in Helixer's predictions, with TAIR12's boundaries aligning more closely to Helixer and supporting the view that newer, long-read-based annotations provide a more reliable benchmark for this aspect of gene structure.

The FASTA files required for the planned Plant-PLMview motif analysis, extracted from the genomic regions spanning -1000 to +500 bp around Helixer-specific predictions and around Helixer-defined 5'UTRs absent from the reference, have already been generated. Completing this analysis would help determine whether these Helixer-specific 5'UTR extensions correspond to regulatory regions.

Use of repetitive and mobile element annotations. Although the TE-overlap analysis was restricted to *Zea mays* in this study, TAIR12 also includes substantially expanded annotation of TE relative to ARAPORT11. Overlaps between Helixer CDS predictions and these newly annotated mobile elements in *A.thaliana* have already been identified during this project but not yet further characterized. Extending the TE-association framework developed for maize to this TAIR12-specific TE annotation would allow a direct comparison of how genome

complexity and repeat content affect Helixer's behavior across a model and a non-model/complex genome.

Toward functional validation beyond sequence similarity. Comparing differentially expressed genes between Helixer and ARAPORT11 would provide an expression-based validation of Helixer-specific predictions, complementing Pfam and UniProt analyses.

Outlook on alternative prediction tools. While this study focused on Helixer, other deep learning-based annotation tools such as Tiberius⁵ have recently been proposed, relying on different architectures and training strategies. Although not currently included in GBOT, the HelixFlor comparison framework developed here could be applied to Tiberius predictions in a similar manner.

Broader relevance. More generally, systematic benchmarking is motivated by the fact that over 83% of the more than 64,850 eukaryotic genomes in the NCBI database still lack gene structure annotation²². In this context, tools such as Helixer provide a scalable solution for first-pass annotation of new or poorly characterized genomes. While performance is strong on model species, the results across this study show that accuracy varies with genome complexity, particularly in repeat-rich regions and at UTR boundaries, highlighting the importance of considering these limitations in non-model applications.

Perspectives for GBOT integration. Finally, beyond the analyses presented here, integrating HelixFlor's comparison outputs directly into the GBOT web interface would allow users to visualize, for any given genomic region, how Helixer predictions diverge from the corresponding reference annotation. Such an integration would extend GBOT's existing comparative genomics tools to the direct evaluation of automatic annotation methods, making the results of this project usable beyond the scope of this internship.

CONCLUSION

This internship extended a previous GBOT evaluation of Helixer to *Zea mays* and *Arabidopsis lyrata*, while also examining the impact of reference annotation choice through a comparison of the ARAPORT11 and TAIR12 annotations of *Arabidopsis thaliana*.

Overall, Helixer produced structurally coherent gene models across diverse plant genomes and maintained strong agreement with expert-curated annotations at the CDS level. The *Arabidopsis lyrata* analysis showed that Helixer can perform well in species absent from its training set, whereas the maize analysis highlighted the challenges posed by repeat-rich genomes, where some Helixer-specific predictions overlapped transposable elements.

The comparison between ARAPORT11 and TAIR12 further demonstrated that differences observed during benchmarking do not solely reflect prediction performance, but can also arise from variation between reference annotations. This highlights the importance of evaluating annotation tools against multiple references whenever possible.

This work also contributed to the development of the HelixFlor comparison framework, providing a reproducible approach for comparing genome annotations at both structural and functional levels. Future work could extend these analyses to additional annotation tools and incorporate complementary sources of evidence, such as transcriptomic data, to further improve the evaluation of automated gene predictions.

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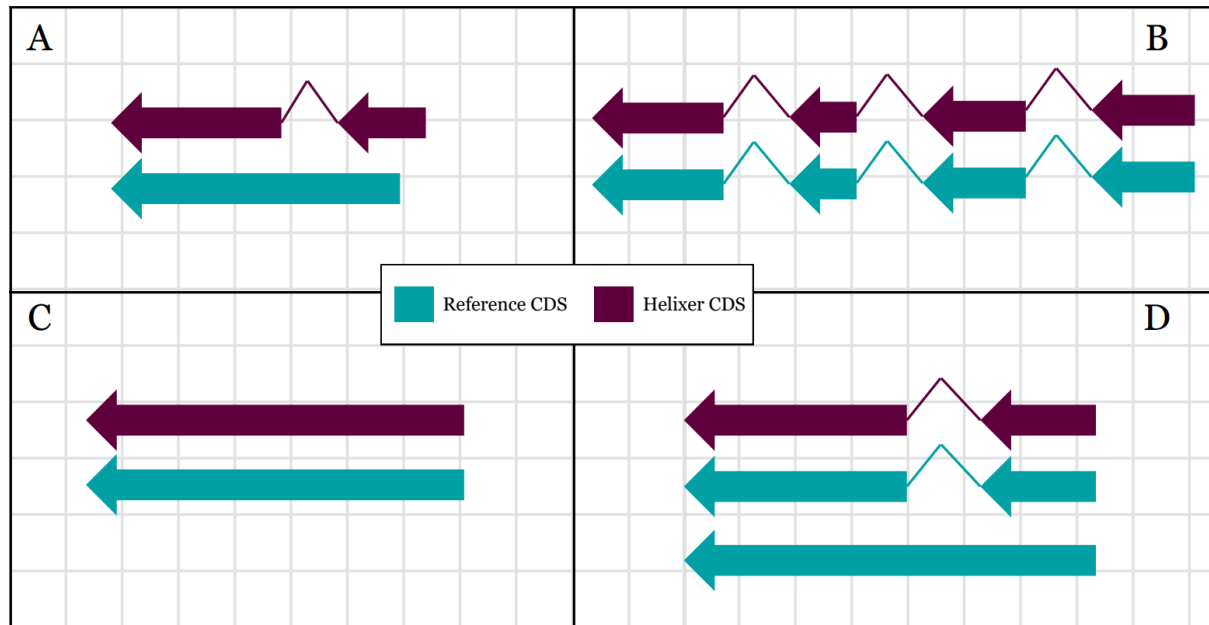
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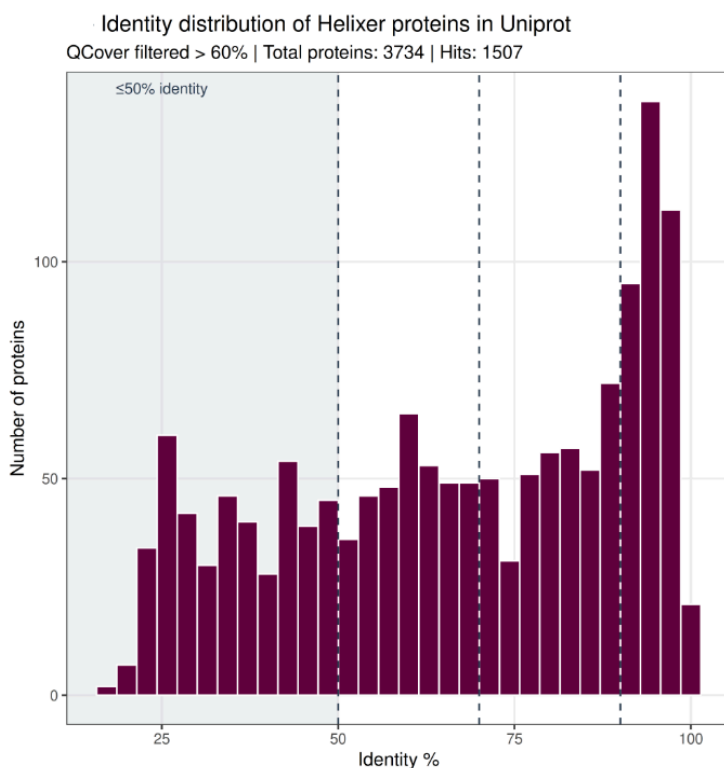
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APPENDIX



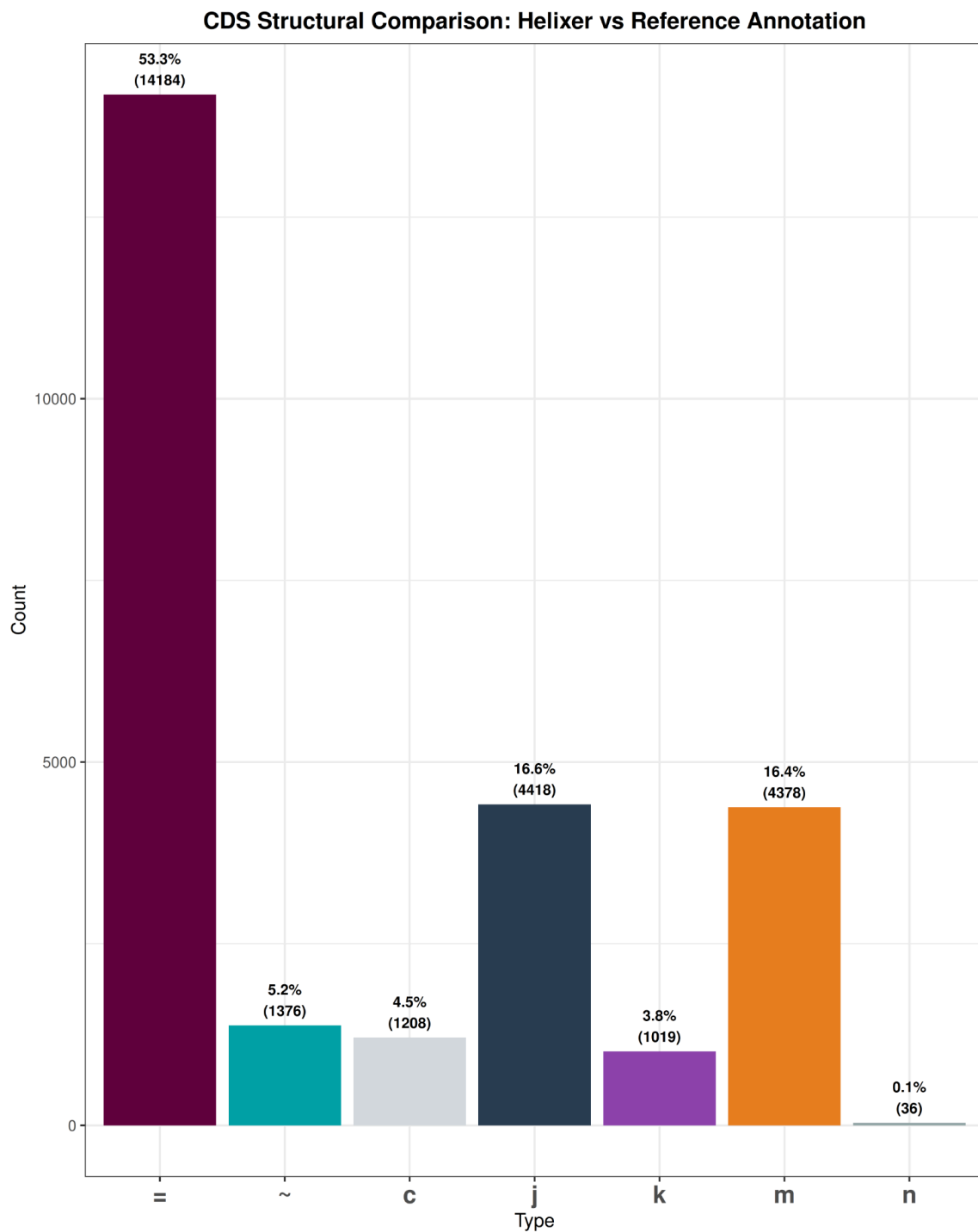
Supplementary Figure 1: Detailed classification of CDS overlap configurations between Helixer and reference annotations.

Schematic representation of additional overlap scenarios between Helixer and reference CDS not covered in Figure 3, including cases of identical length, partial overlap, and nested/contained features. This classification supports the overlap-detection logic used by HelixFlor to assign exon-level identity scores.



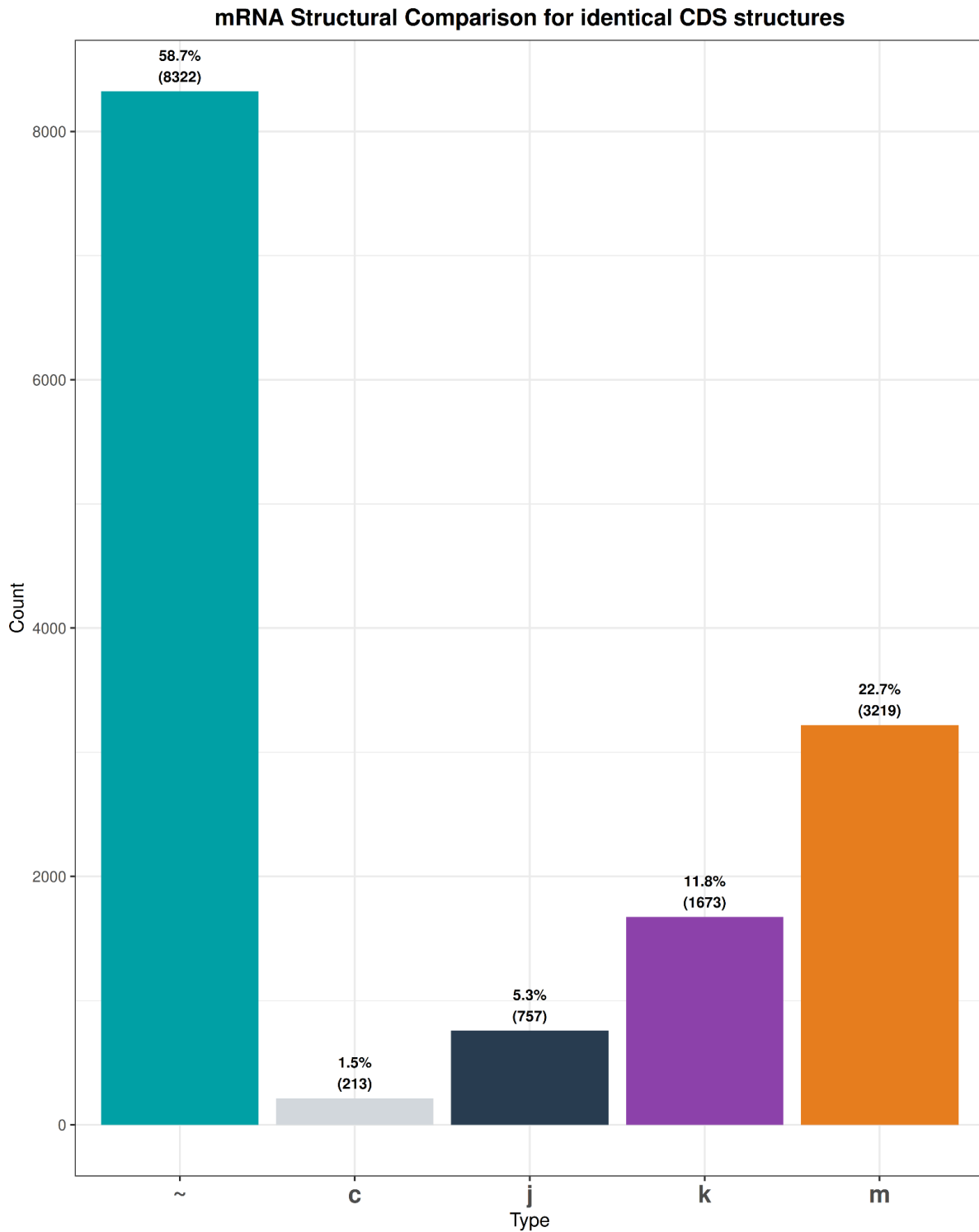
Supplementary Figure 2: :
Sequence identity distribution of
Helixer-specific (Added) proteins
against UniProtKB in *A. lyrata*.

*Distribution of BLASTP identity percentages for the 3,734 Helixer-specific ("Added") proteins in *A. lyrata*, filtered at query coverage > 60% (n = 1,507 hits). Dashed lines indicate 50%, 70%, and 90% identity thresholds.*



Supplementary Figure 3A: CDS structural agreement categories between Helixer and the reference annotation in *A. lyrata*.

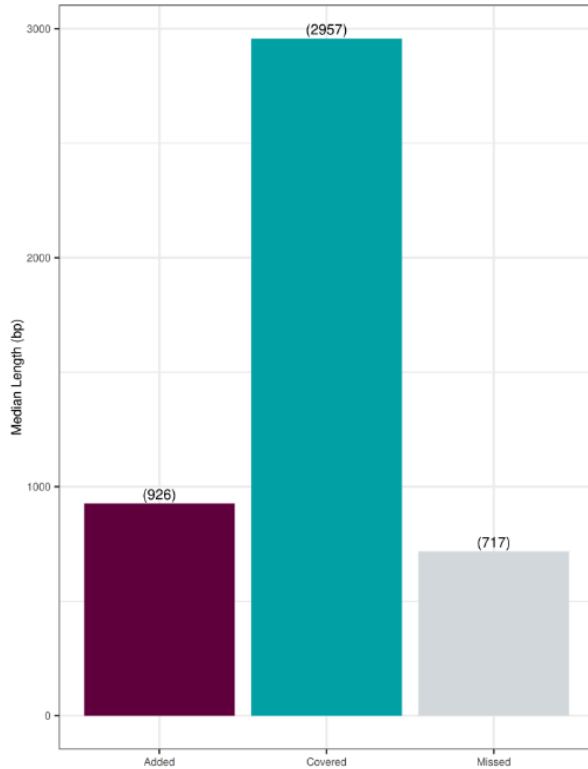
Number of gene models per structural equivalence category ("=", "~", "c", "j", "k", "m", "n", as defined by the E2annot classification in Figure 5) for CDS comparisons between Helixer and the *A. lyrata* reference annotation.



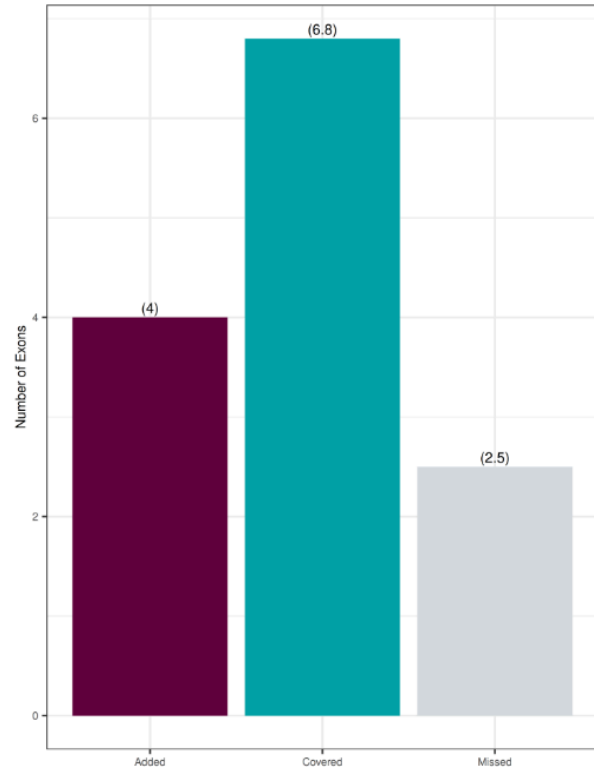
Supplementary Figure 3B: mRNA structural agreement categories for transcripts with identical CDS in *A. lyrata*.

Distribution of structural equivalence categories at the mRNA level, restricted to transcripts with identical CDS structures, illustrating that most UTR-level differences fall into the "~" (conserved intron chain) category.

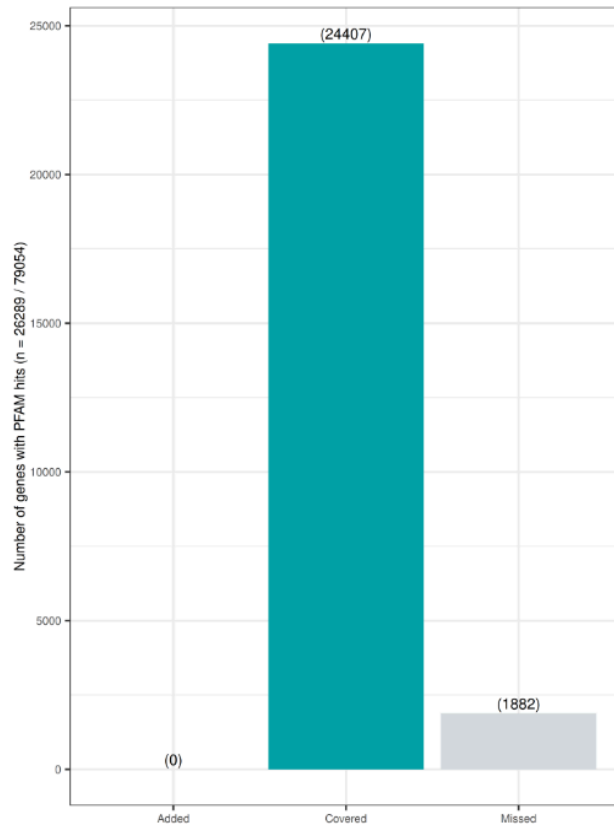
A - Median Length of Helixer CDS | B73
Added genes : 7267 | Missed genes : 7081 | Common genes : 64706



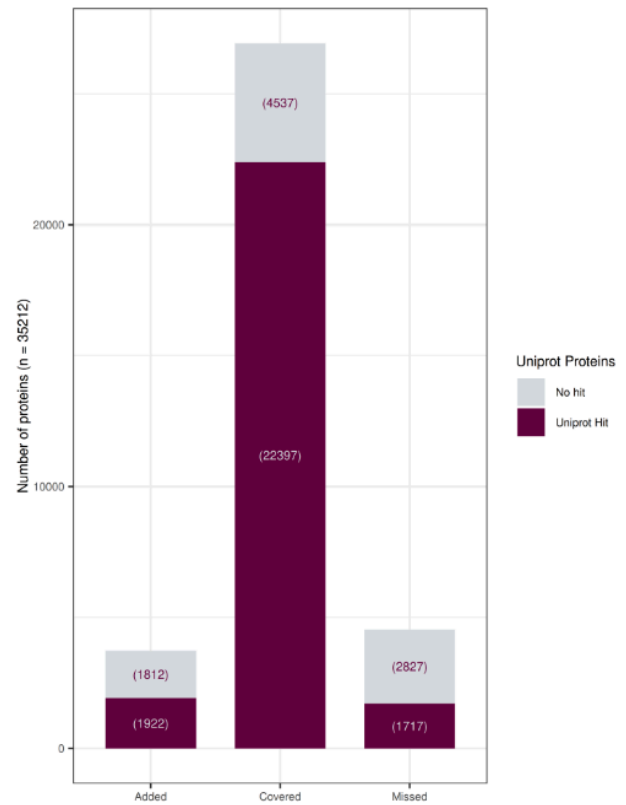
B - Mean Exons Number in Helixer CDS | B73
Added genes : 7267 | Missed genes : 7081 | Common genes : 64706



C - Distribution of Genes with PFAM hit(s) in GBOT | B73



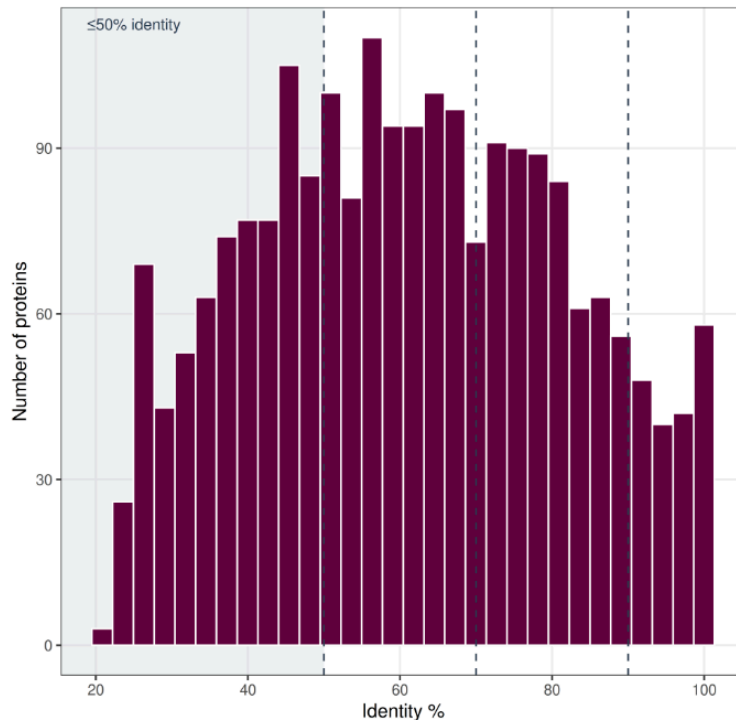
D - Identification of Helixer proteins in Uniprot | B73
Identity $\geq$ 0% | Qcover $\geq$ 0% | Evalue $\leq$ 1e-05



Supplementary Figure 4: Structural and functional characterization of Helixer-predicted gene categories in *Zea mays* (B73).

Comparative analysis of genes classified as Added, Covered, and Missed in *Zea mays* B73. (A) Median CDS length per category. (B) Mean exon number per category. (C) Number of genes with at least one PFAM domain hit. (D) UniProtKB match status of Helixer-predicted proteins (Identity $\geq 0\%$, QCover $\geq 0\%$, E-value $\leq 1e-05$).

A - Identity distribution of Helixer proteins in Uniprot
QCover filtered > 60% | Total proteins: 7267 | Hits: 2146



Supplementary Figure 5: UniProt identity distribution of Helixer-specific proteins in *Zea mays* (B73).

Distribution of BLASTP identity percentages for the 7,267 Helixer-specific proteins in *Zea mays* B73, filtered at query coverage > 60% ($n = 2,146$ hits), with thresholds marked at 50%, 70%, and 90% identity.

B73: Genome annotations summary

mRNAs Structural analysis for specific CDS structures

CDS Structure	Number	Mean Identity %	Most Frequent Type	5'UTR Helixer Length	3'UTR Helixer Length	5'UTR Reference Length	3'UTR Reference Length
Identical	18250	84 ¹	~	474 ^{1,2}	348 ^{1,2}	431 ^{1,2}	485 ^{1,2}
Almost Identical	1434	81 ¹	~	437 ^{1,2}	301 ^{1,2}	394 ^{1,2}	377 ^{1,2}

¹ Only mRNAs linked to the CDS above were compared here.

² Average Length of UTR regions

Supplementary Table 1: UTR structural comparison between Helixer and the reference annotation in *Zea mays* (B73).

Mean 5' and 3' UTR lengths and sequence identity for transcripts with Identical ($n = 18,250$) and Almost Identical ($n = 1,434$) CDS structures between Helixer and the B73 reference annotation.

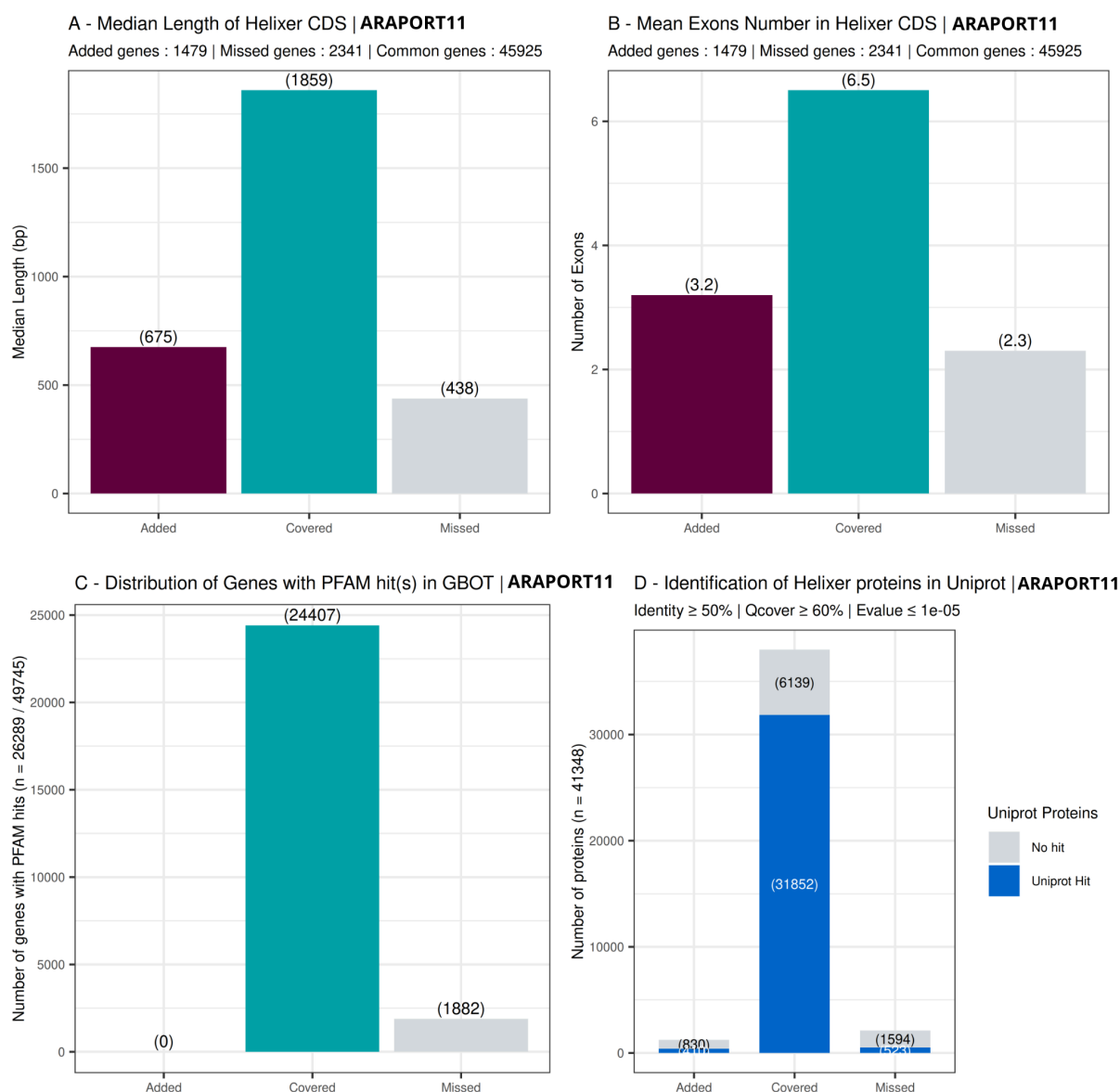
TAIR12: Genome annotations summary

TAIR12 VS ARAPORT11 : Protein Comparison

Shared Proteins	Count
With Same Sequence	22,591
With Different Sequence	7,852
Total	30,443

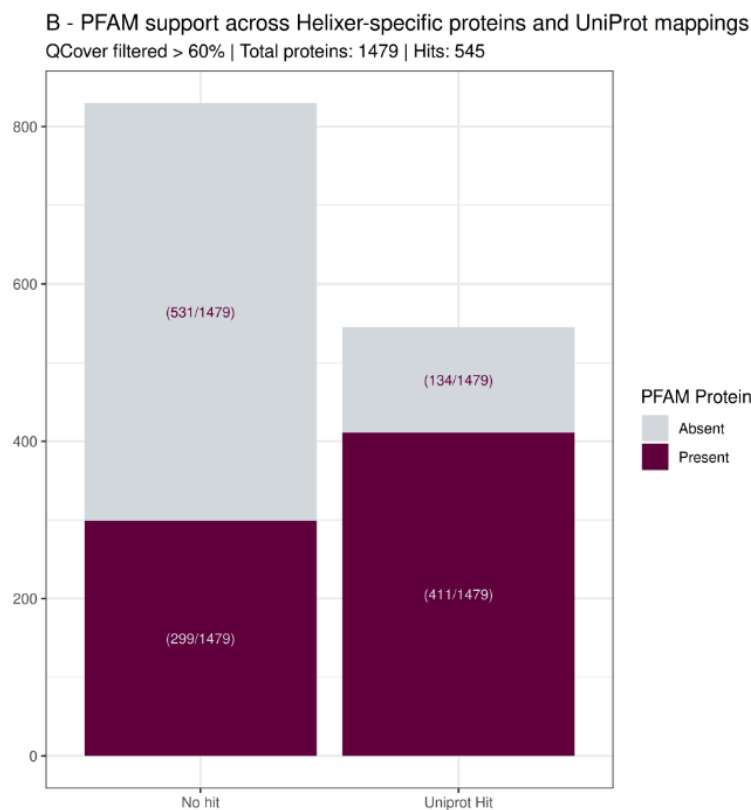
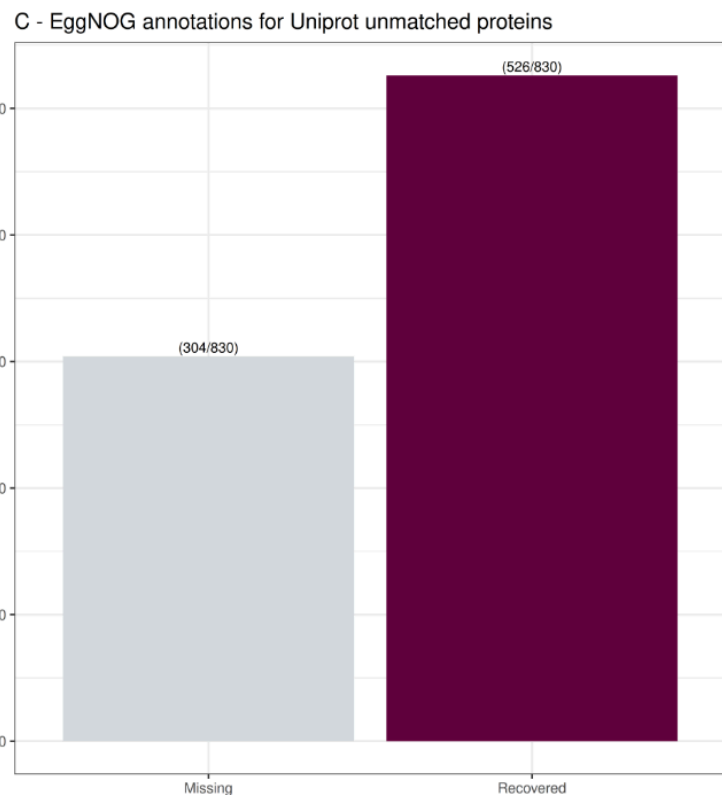
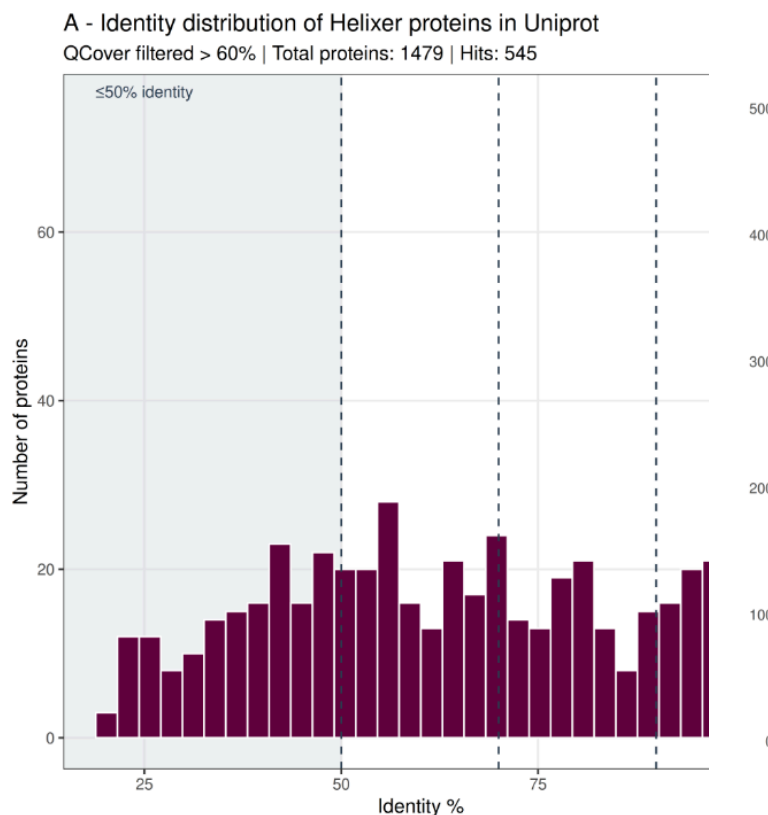
Supplementary Table 2: Protein sequence conservation between shared TAIR12 and ARAPORT11 loci.

Number of shared protein-coding loci ($n = 30,443$) between TAIR12 and ARAPORT11 encoding identical ("With Same Sequence", $n = 22,591$) versus divergent ("With Different Sequence", $n = 7,852$) amino acid sequences.



Supplementary Figure 6: Structural and functional characterization of Helixer-predicted gene categories in ARAPORT11.

Comparative analysis of genes classified as Added ($n = 1,479$), Covered ($n = 45,925$), and Missed ($n = 2,341$) relative to ARAPORT11. (A) Median CDS length. (B) Mean exon number. (C) Number of genes with a PFAM domain hit. (D) UniProtKB match status of Helixer proteins (Identity $\geq 50\%$, QCover $\geq 60\%$, E-value $\leq 1e-05$).



Supplementary Figure 7: UniProt and functional support for Helixer-specific (Added) proteins relative to ARAPORT11.

(A) UniProt identity distribution for Added proteins with a significant hit (QCover > 60%; total = 1,479, hits = 545). (B) PFAM domain support stratified by UniProt hit status. (C) EggNOG functional annotation recovery for UniProt-unmatched Added proteins ($n = 830$): Recovered ($n = 526$) vs. Missing ($n = 304$).

ARAPORT11: Genome annotations summary

mRNAs Structural analysis for specific CDS structures

CDS Structure	Number	Mean Identity %	Most Frequent Type	5'UTR Helixer Length	3'UTR Helixer Length	5'UTR Reference Length	3'UTR Reference Length
Identical	20309	88 ¹	~	241 ^{1,2}	207 ^{1,2}	255 ^{1,2}	276 ^{1,2}
Almost Identical	410	82 ¹	~	224 ^{1,2}	148 ^{1,2}	121 ^{1,2}	178 ^{1,2}

¹ Only mRNAs linked to the CDS above were compared here.

² Average Length of UTR regions

Supplementary Table 3: UTR structural comparison between Helixer and the reference annotation in ARAPORT11.

Mean 5' and 3' UTR lengths and sequence identity for transcripts with Identical (n = 20,309) and Almost Identical (n = 410) CDS structures between Helixer and the ARAPORT11 reference annotation.